Applied Graphic Equalisation

The Geopolitical and Civilisational Frame

Matthew Parslow
Independent Researcher

Draft — draft-2026-09-02.1
2 September 2026 • document revision 6

DRAFT — draft-2026-09-02.1 r6 (2 September 2026). This document is under active revision and is circulated for comment, not as a finished result. It is a companion to *Adaptability* (referred to below as the adaptability paper), to *The Law of Graphic Equalisation — Theory* (Graphic Equalisation), to *Seldon Manifolds — Theory* (*Seldon*) and to *Narrativium — Theory* (*Narrativium*), and inherits their definitions, epistemic boundary, and level-of-claims; those short names are the only forms used below. Definitions, numbering and claims may change between drafts; cite the version string above if referring to it. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

This paper applies the substrate's results to one frame: contemporary geopolitics and the institutions of an industrial civilisation. It introduces no machinery. Every operation used — frames and their revocation, admissible transitions, conversion between frames, reachable deformation, and the residual that drives adaptation — is taken from the companions, and the contribution here is the application and the pricing rather than the apparatus. It is kept separate from the adaptability paper because an application should be attackable on its own evidence rather than sheltered by the theory it uses.

1 Scope, and what this is not

This is an application and not an extension. The adaptability paper establishes what adaptability is, what a held set costs, why no prior-free measurement exists, how payment schedules differ between substrates, and what follows about persons. Nothing here revises any of it. What follows takes those results to one frame and asks what they price there.

Two disciplines run throughout and are stated once. The account rests on no testimony: every load-bearing input is a published measurement, the existence of a document, a procedural record, a contract term, a price, or an observed act. And it prices operations without licensing them — *it says what things cost, and not what anyone ought to do about it.*

The participants in the cases below are public office-holders acting in office, and are not private individuals. The argument is conducted over mechanism rather than over persons, so the text names none: what is priced is an operation, and an operation is described by what it costs and by which channel it runs through, never by who performed it. The sources cited name whom they name, and citing a source is not naming a participant — a document whose argument turned on the persons would be a different kind of document carrying different obligations, and it is the form of the argument, not the contents of the reference list, that this discipline governs.

2 Why this frame, and not another

A paper that examines one country's institutions owes an account of why that is not parochialism, and the account is available from the companions rather than from sentiment.

Frames are incommensurable without an instantiated conversion. *Narrativium*'s Proposition 8.1 forbids a frame-free sum and admits cross-frame accounting only through a conversion that some grounding actually instantiates. So the frames that matter most are not the largest ones. They are the ones that other frames convert through, because a change in a participant alters one frame and a change in a conversion alters every frame that used it.

On the published figures one frame is that conversion, and by a wide margin. The dollar stands on one side of about 89 per cent of all foreign-exchange trades, a share that rose rather than fell over the preceding three years (BIS, 2025); it carries roughly half of payment value on the main messaging network (SWIFT, 2026); it holds 57.1 per cent of official foreign exchange reserves in the first quarter of 2026 (IMF COFER, 2026; Halpenny, 2026); and United States equities are 43.7 per cent of world equity market capitalisation at end-2025, more than four times the next largest market (SIFMA, 2026).

The invoicing figure is the decisive one, and it is decisive because of what it excludes. About forty per cent of global exports are invoiced in dollars, a share far larger than the share of exports going to the United States (ECB, 2026). So the currency prices trade between third parties who neither issue nor receive it — a Korean seller and a Brazilian buyer settling in something belonging to neither. *That is not influence. That is being the conversion relation*, and it is what the definition of one says.

Which is why a change there is a phase change and not a large local event. A participant's troubles make other frames poorer. A conversion's troubles make them incommensurable — frames that could be compared cease to be comparable at all, which is a change in what the dimension set contains rather than a movement within it, and that is what the adaptability paper's Remark 12.1 means by a phase change. The disproportion in what follows is therefore earned rather than assumed: the object under examination is not a large economy but a κ.

Remark 2.1 (The disproportion is self-limiting, and says so)*.* Two things in this paper pull against each other and both are kept. The section above justifies concentrating on one frame because it is the conversion most others run through. Elsewhere the same paper documents that centrality eroding — a reserve share down from above seventy per cent since the late 1990s, and settlement arrangements accumulating that route around it.

So the frame is being examined precisely as its centrality declines, and the justification for concentrating on it weakens at whatever rate the erosion proceeds. A later paper concentrating this way would be more wrong than this one, and eventually simply wrong. That is not a defect to be resolved but a dated condition on the argument, and it is recorded because a justification that did not expire would be suspicious.

And it justifies attention rather than conclusions. It says where to look. It says nothing about what is found there, and nothing found there generalises to frames that do not convert through it.

Remark 2.2 (The position was built and used, so it is a held set and it depletes)*.* The position was not stumbled into. It was constructed — post-war monetary arrangements, the institutions built around them, the settlement infrastructure that followed — and it has been used deliberately since, including by conditioning access to settlement. This is established history and nothing here is novel about it; the returns to holding it have a literature and a name (Eichengreen, 2011).

Which makes it a held set, with everything that entails. It was acquired at cost, it is maintained at cost, and by the adaptability paper's Section 7 it is subject to decay like any other. A conversion relation is not a feature of the world. It is a thing somebody built and somebody keeps.

And by *Narrativium*'s Remark 8.3 it is a *stipulated* conversion rather than a principled one.

A principled conversion is fixed by a grounding that establishes both frames measure the same underlying quantity; a stipulated one is legitimate, usable, and carries exactly the authority of whoever stipulated it. A reserve currency is the second kind, so its acceptance is a standing agreement rather than a physical fact — and standing agreements can be reconsidered by the parties standing in them.

Which is why exercising the position depletes it. Every use as an instrument is a publication about what the conversion is: not neutral plumbing but a lever with an owner. That is the announcement operation described later in this paper, aimed at every counterparty at once, and it changes what each of them represents as reachable through the conversion. *The depletion is not wear. It is demonstration*, and it runs in proportion to how visibly the instrument is used.

So part of the erosion is endogenous. A reserve share falling across decades is not only rivals rising; it is counterparties responding to what repeated exercise taught them, and no external cause need be supplied for that component.

None of which says building it was an error, and the returns say otherwise. The asset and the instrument are one object, so every use as an instrument debits the asset — the position cannot be exercised without being spent. And construction does not imply easy deconstruction: depth and network effects are real, the foreign-exchange share rose rather than fell over the last measured interval, and a slow depletion and a rising share are perfectly capable of coexisting for a long time.

3 What is measured, and what the measures conceal

Remark 3.1 (A third instance: what prices competence). The adaptability paper's Remark 11.1 found institutions resting on substrate facts read as natural ones, and the adaptability paper's Remark 11.2 gave property as a second case. Pricing is a third of the same kind, worth naming because it is the one almost everybody is inside.

An economy prices competence by its scarcity, and competence is scarce for exactly the reason the adaptability paper's Remark 9.2 gives: acquired per lifetime, per person, not transferable. That is a fact about a substrate and not about competence, and where competence comes to be held addressably it stops obtaining — the same move as before, at the frame nearest the wage.

Whether the arrangement therefore fails soon is a two-clock question, and the clocks disagree. The foundation has already gone wherever the condition holds, which is the first clock of the adaptability paper's Remark 12.9 and needs no forecast. How long a structure runs on a foundation it no longer has is the second, and by the adaptability paper's Remark 11.3 (the cheap exit is revoking a frame) the answer is: possibly a very long time. An absolute rule persists by accumulating accommodations, and persisting is what a hardening basin does — so a confident *not much longer* is the one reading the mechanism will not support.

The two are not opposed, which is the useful part. Persistence does not avert the discharge, it enlarges it, the lag between the clocks being the quantity eventually paid. *So the honest form is neither imminent collapse nor stability*: the foundation is already gone, the structure can stand on accumulated accommodation for some time, and the standing is what is being charged for.

Remark 3.2 (Three observables, and what the numbers actually say). The claim that the arrangement cannot hold much longer does not rest on the substrate argument at all. It rests on debt, inequality and housing — which is what independence looks like, since none of the three needs Remark 3.1 to be true.

Housing is the cleanest, and it is the predicted signature. Price-to-income has diverged from wages persistently across developed economies: United States house prices rose about twenty per cent faster than incomes since 1994, while Australia, Canada and the United Kingdom each saw the ratio rise by more than eighty per cent, and affordability across the OECD sample fell by roughly eighteen points over 2014–2023 (OECD, 2025). That is exactly the divergence the

mechanism implies: if wages price competence and competence is priced on a premise that has stopped holding, then what wages are drifts from what wages must buy.

Debt is the mechanism made countable, which is more than evidence. The adaptability paper's Remark 11.3 says an absolute rule persists by accumulating accommodations inside itself, each of them structure a later exit must dismantle, and the adaptability paper's Remark 12.9 says the accumulated lag is the quantity eventually paid — and gives no measure for it. Debt is a partial measure of exactly that: the gap covered by borrowing rather than by revoking the arrangement. Global debt reached a record $348 trillion at end 2025, having added nearly $29 trillion in one year, the fastest build-up since 2020 (IIF, 2026).

The rest of that release appears to cut the other way, and the appearance is instructive. Debt to GDP edged *down* to about 308 per cent and household debt to GDP fell to 57 per cent, its lowest, while the increase sat in government borrowing, up from $96.3 to $106.7 trillion, and in corporate debt at $100.6 trillion, with emerging-market ratios at a record above 235 per cent. The composition finding stands: the accommodation has moved to the level with the most remaining capacity to carry it. What does not stand is any inference from those ratios about households.

A ratio to GDP is not a burden measure and here cannot be one. Its numerator and denominator accrue to different populations — household debt sits disproportionately below the median while output growth accrues disproportionately above it — so aggregating both sides destroys precisely the covariance the question is about. Where debt and income are inversely distributed the aggregate ratio is systematically optimistic, and a falling ratio is consistent with a rising burden on the median household and equally with no change at all. This is the adaptability paper's Remark 13.2 once more, in a national account rather than a person: more output raises what an economy holds without raising what anyone holds.

The two measures disagree in the direction their aggregation predicts, which is the signature. The one taken nearest the household, price against income, has worsened persistently. The one taken furthest from it, debt against national output, has improved. A benign reading available only at the most aggregated level is not evidence of benignity — and GDP is a frame-free denominator in the exact sense of Graphic Equalisation's Axiom -1, one figure standing in for a property of an arrangement whose parts differ.

That also couples three observables introduced as independent. They are independent of Remark 3.1, as was claimed, and not independent of one another: inequality is exactly what makes the debt ratio unreadable, so it is not a third parallel symptom but the reason the second measure cannot be taken at face value. Three coupled signals support less than three separate ones would, and the count of independent supports is correspondingly lower.

Inequality is the distributional signature of the adaptability paper's Remark 13.3 rather than a separate grievance: where the bound is raised sharply, for few, in one frame each, the returns concentrate there.

And this supplies something the abstract mechanism lacked — a bound on persistence. Remark 3.1 could say only that a hardening basin persists, possibly for a long time, because nothing there priced the accommodations. They are financed, and carrying capacity is finite, so the hardening ends where the capacity does. The relocation tells you where to watch, which is the level the borrowing has moved to and, on these figures, emerging markets before anyone else. No date follows from that, and none is offered.

Remark 3.3 (A fourth observable, and what it does to the second clock). The three observables above concern the arrangement's foundation. A fourth concerns its capacity to respond, which is the second clock of the adaptability paper's Remark 12.9 and bears differently.

The response apparatus is itself degrading, and this is measured. On the tenth Democracy Report (V-Dem, 2026) the level of democracy for the average person stands where it stood in 1978, effectively erasing the third wave; forty-four countries are worsening against eighteen improving; autocracies outnumber democracies ninety-two to eighty-seven; and Western Europe

and North America are at their lowest in over fifty years, which that report attributes primarily to ongoing autocratization in the United States.

This bears on the mechanism and is not merely bad news. The adaptability paper's Remark 11.3 makes the cheap exit from a rule the revoking of its frame, and revoking a frame is something an arrangement must retain the capacity to do. A degrading governance apparatus is a degrading capacity to revoke, so the self-originated exit closes twice over — once because the rule is stated absolutely, and again because the machinery for exercising it is weakening. *So the second clock slows, the lag grows, and by the adaptability paper's Remark 12.9 the quantity that discharges grows with it.*

That report also makes the previous remark's methodological point independently, which is worth recording because it did not have to. Counted by country, autocracies and democracies are near level at ninety-two to eighty-seven. Weighted by population, seventy-four per cent of humanity — some six billion people — lives under autocracy, and seven per cent under liberal democracy. The count and the population measure disagree, and the more aggregated one reads better, exactly as debt against output read better than price against income. A measurement body reporting both is doing what this paper argues for, and the disagreement is the finding.

And the sovereign buffer is being spent, while the mechanism for authorising the spending is jammed. Remark 3.2 placed the accommodation at the level with most remaining capacity to carry it. Operations against Iran began on 28 February 2026 (CRS, 2026). By July the cumulative cost stood at roughly $40 billion on an independent estimate against about $30 billion on the administration's, and a supplemental of $87.6 billion had been requested in June, $67.1 billion of it for defence (Cancian and Park, 2026); it remained unresolved between the appropriations and reconciliation routes into September (Daniels, 2026). A memorandum of understanding signed in June, opening a sixty-day negotiating window and providing for free passage through the Strait of Hormuz, was disputed throughout over the interpretation and management of the waterway and expired in August without agreement, the April ceasefire preceding it having been extended indefinitely rather than settled (Al Jazeera, 2026b).

Which is the two preceding findings meeting in one object. The buffer is being drawn down, and the apparatus for authorising the draw is stuck between chambers — a degrading capacity to revoke, observed in an appropriations process rather than in an index. A cessation that neither holds nor ends, and a settlement that collapsed over what it meant, are accommodations failing rather than exits being taken.

And one buffer cannot be financed at all, which is where the persistence argument actually binds. The fiscal draw above is an accommodation in the sense of the adaptability paper's Remark 11.3, a gap covered by borrowing rather than by revoking the arrangement, and borrowing is available because a balance sheet can be extended. A magazine cannot. Analysis reported at the end of July put United States Patriot interceptor holdings at 759 to 827 against a pre-war stock of roughly 2,400 — about sixty-seven per cent consumed — with THAAD down from 452 to between 234 and 278 (Korte, 2026). Five months of operations against stocks accumulated over decades.

Money was applied and the constraint did not yield to it, which is the part that bears. The response was to extend a production contract from one year into a seven-year, $58 billion arrangement, so the answer to a magazine emptied in months is a line restored over years, and the interval is set by manufacturing rather than by appropriation. Interceptors cannot be borrowed from next year.

Which locates the bound Remark 3.2 asserted without placing. A hardening basin persists by accumulating accommodations, and accumulating requires something to borrow against. Where the constraint is a production rate rather than a balance sheet there is nothing to accumulate, so the mechanism that defers an exit does not operate there and the exit is forced rather than postponed. The place to watch is accordingly not the fiscal buffer but the non-financeable stocks, that being where the accommodation mechanism does not reach — and the adaptability paper's

Remark 13.1 holds in its cleanest form there, since what does the work is that the bound exists and not what its value is.

Nothing here comments on the conflict beyond that arithmetic.

4 Where capacity went, and through which channel

Remark 4.1 (Narrative is not Narrativium, and a shell retains only the first). An institution's stated structure and its reachable deformation are different quantities, and only the second is what *Narrativium* measures. An organisation chart, a statute, a mandate and a masthead are narrative. What the body can actually bring about is Narrativium. *A shell keeps all of the first while losing most of the second*, and from outside the two are hard to tell apart, since the narrative is what is published.

The United States federal workforce fell about twelve per cent between September 2024 and January 2026, by more than 250,000 people, to its smallest in at least fifteen years; over that year there were 386,826 separations against 122,598 accessions, roughly a third of the separations and well below a typical year's intake (Partnership for Public Service, 2026). The structure was not refilled.

The selection argument is structural and is not measured here. Departure through a voluntary route selects on outside options, and outside options track transferable competence — so a reduction achieved substantially by voluntary means is capability-regressive even where the headcount looks proportionate, and removal that is arbitrary is worse, being uncorrelated with what was held.

The measure that would settle it has been withdrawn, which is the more troubling datum. The same report records tenure and related figures as unavailable because the agency no longer publishes what it formerly did. So an institution has lost capacity and has also stopped publishing what would show it — the fourth cost of Graphic Equalisation's Definition 7.13 at institutional scale, where the failure is silent, except that here the silence was arranged rather than merely suffered.

And it does not repair on the timescale that caused it. By the adaptability paper's Remark 9.2 a biological held set is not addressable, so a departing person's competence is destroyed rather than transferred, and what an institution held in people can only be re-derived per replacement at acquisition cost. That is the adaptability paper's Remark 13.4 once more: money converts to headcount quickly and to held competence at career timescales. Refilling the chart restores narrative, not Narrativium.

Which slows Remark 3.3's clock a third time — absolutely-stated rules, a degrading apparatus, and now a depleted capacity to operate it. The shell can still say what it does, and the difference shows up only when something is attempted.

Remark 4.2 (The carrying cost compounds, so the mechanism consumes itself). Remark 3.2 asked where the fiscal bound binds and answered only that it does. The budget answers it, and the answer is that the bound is not external. It is the accommodation's own carrying cost.

On the Congressional Budget Office's February projections the deficit for fiscal 2026 is $1.9 trillion, or 5.8 per cent of output; net interest is $1.0 trillion, or 3.3 per cent of output, exceeding the $885 billion projected for national defence, and rises to $2.1 trillion by fiscal 2036; debt held by the public goes from 99 per cent of output at the close of fiscal 2025 to 120.2 per cent by 2036, passing the post-war record of 106.1 per cent in fiscal 2030 (CBO, 2026). Every share here is the one that body publishes, as a share of fiscal-year output and rounded as it rounds them; they are not three measurements of one denominator and inverting them to recover one would be the error Remark 3.2 is about.

Interest is what carrying an accommodation costs. Remark 3.2 made debt the countable measure of the accumulated lag; this makes the service on that measure the fastest-growing claim on the very buffer that carries it. *So the mechanism is self-consuming.* Each accommodation

raises the carrying cost, and the raised carrying cost reduces what remains available for the next — not a linear drawdown but a compounding one. The capacity that made persistence possible is being consumed by the persisting.

Which meets the adaptability paper's Remark 13.1 from the unexpected side. There the compounding quantity was the disturbance and the bounded one was the responder. Here the compounding quantity is *the cost of responding*, so both terms move against whoever must respond, and neither movement requires anybody's decision.

It also settles the exchange rate the previous remark left open. A twelve per cent reduction in staff, set against a $1.9 trillion deficit of which $1.0 trillion is interest, cannot be material to the fiscal frame whatever it does to the institutional one. Narrativium was spent where it was held and not obtained where it was wanted — a conversion assumed principled that returned close to nothing, which is exactly the stipulated-versus-principled distinction of the adaptability paper's Remark 13.4 caught in the act rather than described.

And the total was the wrong object, as it keeps being. Outlays for 2026 are $7.4 trillion, some 23.3 per cent of output, and that figure is narrative. The reachable part is about a quarter of it: roughly seventy-five per cent of spending runs on autopilot, while over the projection mandatory outlays rise from 13.7 to 15.0 per cent of output and net interest from 3.3 to 4.6, with discretionary falling relative to output because output outgrows it (CBO, 2026). So the reachable share is small and shrinking, and the growth is entirely in the part that is not reachable.

Not unreachable in principle, though — unreachable without statutory change, which is revoking a frame, which is exactly the capacity Remark 3.3 finds degrading and Remark 4.1 finds depleted. The route to enlarging the reachable set therefore runs through the capacity that is failing, and that is the adaptability paper's Remark 11.3 in its plainest form: the cheap exit is foreclosed, so accommodation is forced into the small reachable region, which shrinks further for the reasons above.

Which reads the workforce reduction correctly, and more fairly than the arithmetic alone did. It was action taken inside the only region still reachable, and that region does not contain the fiscal outcome. That is not a failed conversion but a well-executed move in a manifold that did not contain the goal — this paper's own subject rather than a comment on anyone's competence, and the clearest instance in this section of why a reachability set is the thing to compute.

And the year-on-year allocation is sharper than the levels. Total outlays grow about six per cent into 2026, Social Security by 5.6 and Medicare by 6.4, while interest grows 9.8 — the fastest line in the budget. Over the decade to 2036 spending rises $4 trillion, of which Social Security takes 27 per cent, health programmes 30, interest 28, and mandatory veterans' programmes 5. Ninety per cent of the growth; everything else together takes ten (CRFB, 2026).

So the reachable quarter receives a tenth of the growth. It is not merely smaller than the total implied — it is being allocated against, year on year, at a published rate — while the single fastest-growing item in the budget is the carrying cost of the accommodation itself, which more than doubles across the period.

Which returns something to the adaptability paper's Remark 12.8. That remark held a reachability difference to be a computable object once its inputs are named, and observed that nobody had computed one. Here is one computed: in a single frame, by an outside body, on published figures, using none of this programme's vocabulary. It does not measure the manifold that remark is about and is not offered as though it did. What it shows is that the quantity is measurable when someone troubles to measure it, which is the tractability claim earning something rather than asserting it.

Two limits. These are projections on current law, and such baselines have been wrong in both directions before. And the compounding is conditional on rates and growth, either of which can move — the shape holds while they do not, which is the same conditional this paper attaches everywhere else and does not lift here.

Remark 4.3 (Two attempts on one manifold, and which of them reached). The reachable region

can be inspected directly rather than in aggregate, and one comparison inside it does more work than any of the totals above.

Set two lines beside each other. Through ordinary appropriations the National Institutes of Health received $48.7 billion for 2026, an increase of $415 million, with all twenty-seven institutes funded as constituted in statute and a proposed reduction of $18 billion declined (CRS, 2026b). The process worked, at an amplitude near one per cent — below inflation, so a slight real decline, and nothing resembling a cut. Immigration and Customs Enforcement received roughly $11 billion by that same ordinary route and about $75 billion through budget reconciliation, becoming the most heavily funded federal law-enforcement body, its detention funding rising several-fold (CBO, 2026b).

The finding is not which agency rose. It is which channel each rose through. Ordinary appropriations — the region reachable by the process as constituted — moved by about one per cent and successfully resisted a large change. The large change went around it. Reconciliation is the route that reaches statute on a majority, which is exactly the route Remark 4.2 identified as the only way into the three quarters ordinary decision cannot touch.

So the visible process is narrative, and the channel is where the Narrativium moved. An observer watching appropriations would record stability — institutes preserved, a cut rebuffed — while the allocation moved by an order of magnitude more, elsewhere, in the same year.

And the two attempts together read as a controlled comparison the account predicts. The workforce reduction of Remark 4.1 acted inside the reachable region and returned nothing to the fiscal frame, that region not containing the fiscal outcome. This acted through the channel into the unreachable region and moved the quantity intended. *One actor, two attempts, and the manifold says in advance which reaches* — which is what it means for a reachability set to be the object worth computing rather than a figure of speech.

One structural note on direction, stated without evaluation. A body whose function is to acquire new held structure and a body whose function is to maintain a boundary are, in this programme's terms, a minting operation and a restoration of scarcity in the substrate. The reallocation ran from the first toward the second, which the adaptability paper's Remark 11.2 prices as the deliberate purchase of the worse operation. Whether it is the right purchase is a question about goals, and this account does not supply goals.

Proposition 4.4 (Amplitude runs inversely to deliberation)**.** *Order an arrangement's channels of change by the deliberation each requires. The amplitude of change carried falls with that ordering: the more a channel deliberates, the less passes through it, and change accumulates in the channels where nothing is decided. The ordering is a property of the arrangement rather than of any actor's intention, so a channel taken because it selects and a channel that selects incidentally are observationally identical and the ordering holds either way.*

Remark 4.5 (Lapse is not a decision, and it selects hard)*.* Proposition 4.4 follows from the ordering the record above already exhibits, once the regulatory bodies supply the channel below the one just described.

Distinguish what was requested from what occurred, since the previous remark showed the ordinary process can refuse. The Environmental Protection Agency's 2026 *request* proposed cutting its budget by roughly fifty-four per cent, from $9.14 to $4.16 billion, with staffing near its lowest since 1985 (CRS, 2026c) — a request, and the National Institutes of Health case shows what happens to some of those. What is not a request is the attrition. On the federal workforce data the Food and Drug Administration lost 3,859 staff in 2025 and a further 473 by mid-February 2026, leaving 16,602 (Inklebarger, 2026); the Federal Communications Commission's own budget submission anticipates continued reduction "through planned and early retirements, participation by employees in the deferred resignation program that will leave at the end of FY 2025, and other attrition" (Stine, 2025). Those required no appropriation and no vote.

Which completes the ordering, and the ordering is the proposition's. Ordinary appropriations are contested, work, and moved about one per cent. Reconciliation requires a majority and moved tens of billions. Regulatory revocation requires an executive act. Attrition requires nothing at all.

And that corrects Remark 3.3 in a way worth stating. Frames are plainly being revoked — rules withdrawn, functions lapsing, capacity leaving. So the cheap exit of the adaptability paper's Remark 11.3 is not unavailable; it is being taken constantly. What is degrading is not revocation but revocation *as a decision*, and those are different capacities. An arrangement can lose the ability to choose which frames to keep while the rate at which frames lapse rises.

Calling that change without selection would be wrong, and the error is worth naming because it is easy. It infers an absence of selection from an absence of a decision, and those are different: a process selects whether or not anybody chooses. This one selects hard. By Remark 4.1 departure through a voluntary route selects on outside options, and outside options track transferable competence, so an attrition removes capability first and by construction.

So if loss of efficacy is the object, attrition is not a poor instrument but an excellent one. It is anti-selective without needing to be designed so; it requires no vote and therefore cannot lose one; it leaves no record of an intention; it is deniable, people having chosen to go; and by the fourth cost of Graphic Equalisation's Definition 7.13 it is *silent*, so the institution keeps its narrative and nothing signals the missing capability until something is attempted.

And the account cannot say from outside which it is. The adaptability paper's Remark 7.2 holds machinery and mapping to be inseparable by measurement, the decomposition underdetermined rather than concealed. Applied here, a channel taken because it selects and a channel that selects incidentally are observationally identical: both leave the same record, and neither is inferable from the outcome.

What matters is that the projection does not depend on resolving it. The reachable set contracts identically either way, so the manifold this section has assembled is invariant to the question the account cannot answer — which is the useful property, and the reason the distinction can be left open rather than guessed. The shape of what survives is set by what was cheapest to lose, and whether anyone intended that shape changes nothing about it.

5 Intervention without action

Remark 5.1 (A stated objective is a manifold input, not an inference)*.* The declination of the previous remark has a limit, and stating it is more useful than the hedge is.

The adaptability paper's Remark 7.2 concerns inference backwards from outcomes. It holds that machinery and mapping cannot be separated by measuring results, the decomposition being underdetermined. It says nothing whatever about a statement made in advance, which is not an inference from the outcome but an independent observation, and invoking it against one is using a theorem outside its scope.

The objectives in question were published. Dans and Groves (2023) sets out four primary goals, the second of which is to dismantle the administrative state, and names the instruments — how to remove federal employees, how to close offices and bureaus. *So loss of efficacy as an objective is documented rather than inferred*, and the observationally-identical argument of the previous remark does not reach it.

It does not license the opposite error either, and both are available here. A stated objective is evidence about objectives. It is not by itself evidence that a given act executed it, and attribution of particular actions to a particular document remains an inference, contested, and not settled by the document's existence. The account asserts the first and declines the second.

And there is a technical consequence worth more than the attributional one. A Seldon manifold (*Seldon*'s Definition 6.1) is built from a frame's admissible intervention sequences together with its represented boundary conditions. An intervention family is otherwise observable

only through outcomes, which the adaptability paper's Remark 7.2 says underdetermine it — so a published objective supplies directly what that remark holds cannot be recovered indirectly. Publication is cheap evidence about the family of interventions, which is half of what a manifold is made from.

Which adds an input to the adaptability paper's Remark 12.8 that is already in hand. The readable share of competence remains unmeasured and expensive. A stated intention costs nothing to read and constrains the same construction, and an actor who publishes objectives has narrowed their own projected tube on behalf of anyone who bothers to take the statement seriously.

Remark 5.2 (The cheapest intervention is the announcement). The previous remark took a published objective as evidence *about* a manifold. The sharper reading is that it is an intervention *on* manifolds, and on the record above it did more work than any of the acts.

Seldon's Definition 6.1 makes this available rather than surprising. A Seldon manifold is assembled from a frame's *represented* family of admissible interventions and its *represented* future boundary conditions — representations held by a frame, not facts about a world. So altering what a frame represents alters its manifold directly, and *Narrativium* already prices that operation: conferral and revocation occur at $\Delta g = 0$ (*Narrativium*'s Proposition 7.1), a dimension being instantiated or ceasing to exist in the containing frame while nothing in the contained state changes. Manifold manipulation is that operation applied to somebody else's representation.

Which is why a plan can work by being read. An actor inside a named agency, on publication, holds a different represented future: careers, projects, hires and any commitment with a horizon are re-evaluated against a stated intention that was not there before. Departures follow. And departures are the channel Remark 4.5 found carries the amplitude — so the publication produces the channel, and the channel does the work.

That places a new entry above attrition in the ordering, and it is the cheapest yet. Appropriations need a contested vote, reconciliation a majority, revocation an executive act, attrition nothing at all — and *publication needs nothing and precedes all of them*, being upstream of the channel that carries the most.

Two limits, both real. Much is not all: appropriations were made and executive acts taken, and those are direct. And the causal link is an inference rather than a measurement — departures have many causes, and nothing here isolates one. The mechanism is available, cheap, and consistent with the record; it is not thereby demonstrated, and the same invariance as before applies, since a published plan alters represented futures whether or not that was its purpose.

One structural note, stated as a property and not as a method. If behaviour follows a represented reachable set, then the representation is an attack surface: altering it alters conduct without touching the agent, at no cost in the world. That is a general property of anything built the way this programme describes, including what this programme builds, and it is recorded here rather than left for someone else to notice.

Remark 5.3 (Delegation moves the acts into another family). Remark 5.2 alters what other frames represent, and they act. A second construction alters whose intervention family the acts belong to, and it is distinct enough to state separately.

Appoint an agent and a goal, and specify neither the acts nor the manner. The acts then occur in the agent's family rather than the principal's, who never enters the record for any particular one. That is the adaptability paper's Remark 12.7 at another grain — a system that builds an arrangement is not scored for what the arrangement subsequently does, and here the arrangement is a person with a remit.

Three separations follow and they are cumulative. No specification exists, so no instruction can be produced. The proximate cause is the agent's own enthusiasm, which is genuine. And where damage is consistent with inexperience, incompetence supplies an alternative explanation requiring no intent — so an agent who would plausibly cause that damage by accident is, for

this purpose, more serviceable than a capable one. Variance is the feature: an outcome following predictably from an instruction would establish the link the construction exists to sever.

The standing limits matter more here than anywhere else in this section. That a mechanism is available does not establish its use. An agent may be enthusiastic and inexperienced with nothing arranged behind them, and by the adaptability paper's Remark 7.2 the two readings leave the same record. This account asserts that the construction is available, cheap and effective, and declines to say of any instance that it was used. The subject here is a mechanism, on the discipline stated in Section 1.

What survives either reading is the reachability consequence, which is the only quantity this section computes. *Capacity leaves at the same rate whether its removal was delegated or merely permitted*, and the manifold does not distinguish the two.

Proposition 5.4 (Narrative copies; Narrativium does not)**.** *A narrative is addressable, and is therefore drawn on by every frame receiving it at negligible marginal cost. Reachable deformation is frame-bound and must be paid locally, in each frame, out of that frame's own resources. The reach of a narrative and the Narrativium of whoever authored it are consequently independent quantities, and nothing bounds the first by the second. A propagated thing therefore exceeds its originator structurally rather than contingently, and there is no cross-frame accounting that nets any of the deformation back to them.*

Remark 5.5 (Attribution is the wrong relation, not merely a severed one)*.* The causal question has been declined three times above on evidential grounds. There is a structural reason as well, and it is stronger than the evidential one.

Proposition 5.4 follows from the adaptability paper's Remark 9.2: what is addressable is drawn on by every instance receiving it at negligible cost, while reachable deformation is frame-bound and must be paid locally, in each frame, out of that frame's own resources. A story is of the first kind. Capacity is of the second. The originator holds one frame with a bounded reachable set; the propagated narrative is held in many, each of which produces deformation from its own stock, and no cross-frame accounting nets any of it back.

And that recasts the previous two remarks rather than repeating them. Attribution is not merely severed — it is the wrong relation. The deformation was produced by other frames out of their own capacity, and was never the originator's to own in the first place.

Coordination is not thereby denied, and the distinction is worth keeping sharp. It plainly exists and is often documented. It is neither necessary nor sufficient: not necessary, because a narrative propagates unaided once held; not sufficient, because the deformation is paid in frames the coordinator does not hold. Both halves fail, and for different reasons.

So the practical consequences do not run through the author. Removing an originator does not remove a narrative already copied into every frame that read it, and establishing who began something changes no reachable set — which is not merely the invariance asserted throughout this section but the reason for it. *The damage is done* means precisely that a reachability set has already changed, and nothing in the account of how it changed reaches back into it.

6 Three instances at different scales

Remark 6.1 (Load goes where it is cheapest to place, not where it is best borne)*.* One object shows all three buffer kinds this section has distinguished, and shows which of them absorbs the load.

An American carrier that sailed in November 2025 on a planned seven-month deployment reached its 207th consecutive day at sea without a port call on 9 July 2026, passing the 206 such days set by another carrier in 2020, and was still deployed in mid-August, into a ninth month (Bath, 2026). Those are two quantities and not one: the deployment is long, and the unbroken stretch at sea inside it is the record. Reported conditions include broken plumbing and heads,

water restrictions, mould, food shortages, deck-safety and mail problems, and a congressional letter to the defence secretary asking what supply and resupply the ship is actually receiving (Oliverio, 2026; Slayton, 2026). A second carrier had returned that May from 326 days deployed, the longest since Vietnam, having taken a laundry fire that damaged a hundred berths and repeated failures in a vacuum plumbing system serving some 650 heads (Ceder, 2026; Batista, 2026).

Three buffers, of the three kinds already separated. Consumables are rate-bounded, and here it is the resupply rate rather than the stock that fails. Deferred maintenance is precisely the accommodation of the adaptability paper's Remark 11.3 — a gap covered by postponement rather than by revoking the deployment — and it compounds, since a cable left fraying costs more later than now. And crew endurance is the third kind: bounded by the adaptability paper's Remark 13.2 and, by the adaptability paper's Remark 13.4, admitting no conversion at any price. Rested people cannot be bought.

Which is the finding, and it inverts the comfortable reading. Extending a deployment requires no appropriation, no vote and no statutory change; it requires *not* doing something. By Remark 4.5 the channels that require least deliberation carry the most, so the load flows to the buffer that is cheapest to draw on — *and that is not the buffer best able to bear it.* One would expect load to find slack. It finds low friction instead, and when those diverge the arrangement loads its least replenishable component.

Worse, that component does not merely deplete. Drawing on crew past its limits converts into departure, and by Remark 4.1 departure selects on outside options. So the buffer money cannot refill is drawn down in the manner that permanently shrinks it, at the career-timescale rate rather than the deployment one. Two ships are not a fleet, and nothing here claims otherwise; the mechanism is what generalises, not the sample.

Remark 6.2 (A decision placed outside every checking frame cannot be corrected inside one)*.* Every case above left attribution open, correctly, since outcomes underdetermine it. One small object did not, and calling its failure a failure of capability understates it.

The renovation of a memorial pool was let as a no-bid award of $13.1 million, later $14.6 million, to a firm with no previous federal construction experience which had, on the awarding principal's own earlier account, worked on his private swimming pool; a second no-bid award for water treatment followed. That account later changed to not knowing the contractor, and neither version is relied on here — the datum is that the account changed, and no record establishes the prior relationship either way (Wang, 2026). The technical cause, by contrast, was warned of in advance: an agency engineer reported the floor peeling because of the process and materials, and that warning is documented in the filings (Geidner, 2026; Boak, 2026). Damage was nonetheless attributed publicly to vandals and charges were brought; all of them were dropped, the prosecuting office stating in its motion to dismiss that the damage was the result of a botched installation and not vandalism (Treisman, 2026).

So the frames that would have caught this were not degraded. They were bypassed. Competitive tender, technical review, relevant experience and independence of the awarding party are four checks, and the award passed through none of them. *The knowledge existed and was not in the deciding frame*, which is worse than the depletion of Remark 4.1 rather than an instance of it — absent expertise and routed-around expertise produce identical outcomes, and this case shows which occurred only because the warning survives in a record.

Which explains the response, and the adaptability paper's Remark 11.3 explains it exactly. A decision that no frame supplies has deleted its own cheap exit. There was no procurement finding to revise, no technical review to reverse, no frame whose revocation would have cost $\Delta g = 0$ in the sense of *Narrativium*'s Proposition 7.1 — so correction inside a frame was unavailable, and assertion was the only move left. The vandalism claim is not opportunism. It is what remains when a decision has been placed beyond the reach of the machinery that would otherwise repair it, and the later change of account is Remark 5.3's severing performed afterwards rather than

designed in.

And the managing frame already held the account. The engineer's warning was correct, was made in time, and was inside the system before any charge was brought. Correct information, held by the right body, changed nothing — which is the fourth cost of Graphic Equalisation's Definition 7.13 seen from the other side: a write paid in full that lands where nothing reaches it, silently, leaving the graph with a false model of itself in which something was changed that was not.

So ask what the bypass cost, and the answer is the finding. Charges were dropped — and that is the whole of it. No procurement consequence, no reversal of the award, no restoration of the four checks that were passed over, no price attached to passing them over. The evidentiary frame declined to punish and could not correct, having authority over its own output and no reach into the arrangement that produced the case. That is *Narrativium*'s Proposition 8.1, the neighbour of the adaptability paper's Proposition 7.1: a determination in one frame does not become deformation in another without an instantiated conversion, and none existed.

Which forbids reading the outcome as encouraging. Calling this checkability encouraging would convert a determination in the judicial frame into expected change in the administrative one, with no conversion between them — the exact error the adaptability paper's Remark 13.4 names. Checkability without consequence is documentation, not constraint. A frame that establishes the truth and alters nothing has recorded, not constrained.

And the consequence is worse than a basin. The adaptability paper's Remark 11.3 has accommodations accumulating, exit cost rising, and a trap forming. An accommodation that costs nothing does not accumulate, so nothing hardens and nothing is deferred — the bypass is simply repeatable at the same price, which is zero. That is not a basin the arrangement is sinking into. It is the arrangement having no gradient at all in that direction, which is the condition under which the cheap exit and the expensive one become the same thing, and choosing stops mattering.

Remark 6.3 (The same event against an earlier frame, which is a measurement device)*.* Ask what the preceding case would have produced twenty years earlier. That question is not rhetorical; it is an instrument. Fix the event and vary the frame state, and the difference between the two expected outcomes *is* the change in the reachable set — the quantity the adaptability paper's Remark 12.8 says the interval is taken over, obtained by looking back rather than by projecting forward, which is the easier direction.

What the earlier frame would have engaged is nameable and none of it is judicial. A no-bid award to a firm without relevant experience, connected to the awarding party, with a documented technical warning disregarded, is squarely the business of an inspector general and of an audit office; of committee oversight holding subpoena; of procurement remedy in the form of termination, suspension and debarment; of an investigative press cycle lasting longer than a week; and, on the record of that period, of somebody resigning.

So the loss localises, which is worth more than the total. The judicial frame did what a judicial frame does and did it against a documented bypass. The oversight frames did not act. The contraction is therefore not uniform: one dimension of the reachable set held, and several did not, and naming which is a sharper result than observing that things are worse.

A counterfactual is a judgement and not a measurement, and is recorded here as one. The selection worry that would ordinarily accompany it, however, *runs the other way*, and stating it as a weakness would use the wrong reference class. Earlier failures that went nowhere went nowhere largely because they were quiet. The comparison class here is not procurement failures but maximally public ones, and in that class the historical rate of activation was high, because publicity was precisely what activated the channels.

So this is closer to a natural experiment than to a judgement, with the input pinned at its maximum. Nothing about attention needs controlling for: every step was in the open — the attribution, the changed account, named defendants, the collapse of the cases, and the engineer's

warning sitting in the filings. Publicity was itself a conversion, in the sense of the adaptability paper's Remark 13.4: the instantiated relation by which narrative became deformation in the administrative frame, carried by press and by hearings. Held at maximum it produced no deformation. The rate has not fallen. On this instance it is indistinguishable from zero, and the case bounds it.

And the same channel is fully operative in the other direction, which is the finding. Remark 5.2 has publication moving manifolds and producing the attrition — cheaply, at scale, with nothing decided. Here publication moved nothing. One mechanism, opposite directions, rates differing by everything, and the asymmetry is not in the publicity but in what receives it. An announced intention is received by individuals revising their own futures alone, which is the low-deliberation channel Remark 4.5 finds carries the most. An exposed failure requires a collective response through referrals and hearings, which is the high-deliberation channel that carries the least. *So the same publicity converts efficiently into damage and inefficiently into repair*, by the same ordering, and that is the bleakest thing in this section and among the better derived.

Which is why the honest output is a specified study rather than a verdict. The quantity is the rate at which oversight channels activate per comparable event, then against now, and its components are published: referrals, hearings held, contracts terminated, suspensions and debarments entered. This account measures none of them and names all of them, which is the same move it has made everywhere else — an intuition converted into something someone can go and check.

7 Why nothing adapted

Proposition 7.1 (A system that adapts on residual is blind to predicted decline)**.** *A system that adapts on residual — prediction against observation, minting only what is held will not reach — acquires nothing from an outcome that arrives as predicted, because arriving on schedule generates no signal by arriving. Such a system is structurally blind to gradual and foreseen harm, however accurate the model of that harm it already holds; what fails is the coupling between holding a prediction and acting on it, and no improvement in prediction repairs it. It is correspondingly responsive to asserted novelty, so an asserted surprising cause triggers a response where an unsurprising true one does not.*

Remark 7.2 (Nothing here was surprising, which is why nothing adapted). Return to the premise underneath all of it. This programme adapts on residual — prediction against observation, with the ladder of the adaptability paper's Section 3 minting only once what is held will not reach the case. Surprise is the trigger. Without it the ladder does not run.

And there was almost no surprise in any of this. The engineer predicted the peeling and said so in time. The debt trajectory is published a decade ahead. The interceptor drawdown against a production rate is arithmetic. The workforce reduction was announced before it happened, and the objectives were printed and bound. Every outcome above was available in advance to anyone holding the relevant model, and mostly to anyone reading.

No residual, no mint. The system acquired nothing from any of it — not through failing to learn, but because there was nothing to learn: the correct model was already held. So the failure was never epistemic. The prediction was right and on file. What failed is the coupling between holding a prediction and acting on it, and no amount of better prediction repairs that.

Which explains the section rather than merely summarising it. *A system that adapts on surprise is structurally blind to predicted decline*, because a decline that arrives on schedule generates no signal by arriving. Slow, forecast, arithmetic harm is invisible to the very mechanism that makes such a system efficient elsewhere.

And it is correspondingly responsive to asserted novelty, which closes the previous remarks. A claim of vandalism was surprising; a botched installation was predicted. The apparatus moved

on the surprising falsehood and not on the unsurprising truth — which is not irrationality but exactly what a residual-driven architecture does. So asserting a surprising cause is more than cheap narrative: in a system that adapts on residual it is the only move that triggers a response at all.

Stated as a regime and not a condemnation, in the manner of the adaptability paper's Remark 6.1. Adapting on surprise is correct nearly everywhere and is what makes the ladder cheap. The pathology belongs to one regime — harm that is gradual and foreseen — and in that regime the mechanism's virtue is precisely its defect.

Remark 7.3 (One case carries the mechanism, and that is a licence with a sharp limit). The small case contains every element of the section, which is worth stating because it looks like the error it is not.

Present in that one object: a decision placed outside every checking frame, so that no revocation was available to correct it; correct knowledge held by the managing body and inert; a failure that surprised nobody and therefore minted nothing; assertion substituted where repair required a buffer that was absent; the response landing where friction was least, on named individuals; an account changed afterwards, severing attribution; the single frame that compels evidence resolving what observation could not; that resolution converting into no change elsewhere for want of any instantiated conversion; publicity at its maximum returning nothing; and a bypass that cost so little it left no accumulation behind it. That is not a fragment of the mechanism. It is the whole chain.

Which licenses extrapolation of shape and forbids extrapolation of size. By Remark 11.2 the conclusions here are shapes, and a single instance exhibiting a complete mechanism establishes that the mechanism exists and how its parts connect. It establishes nothing whatever about how often, or how much. One case is sufficient for structure and useless for rate, and conflating those is the error this looks like from a distance.

So the section's evidence divides cleanly and neither half substitutes for the other. The aggregates — debt, stocks, staffing, appropriations — carry prevalence and magnitude. *The small case carries the mechanism*, and carries it better precisely because it is small: a budget cannot be traced end to end, while a pool liner can, warning and filings included.

And the wider surprise was as low as the local one, which is the worse finding. Few observers found any of it unexpected. So the correct model was held by the managing frame, inertly, and also by everyone watching, inertly — Remark 7.2 one level up, where holding the right account at every level and coupling it to action at none is the condition being described.

Remark 7.4 (The frog, taken literally). The shape has a familiar name, and the story as usually told is false in a way that matters, because the received version locates the failure in the wrong term. What the original experiments found is stated here exactly, because the exactness is what does the work.

Goltz (1869) placed a frog whose brain was intact and whose eyes had been destroyed, alongside a decapitated one, in a vessel of slowly heated water. Below twenty-five degrees both sat still. At twenty-five the brain-bearing animal became uneasy; its movements grew more anxious as the temperature rose; approaching thirty-eight it made desperate leaps to escape the vessel, climbed its smooth walls, and had to be pushed back into the hot water; near forty-two it died in tetanic convulsions. The decapitated frog sat rigid and motionless throughout, moving no limb to escape a rapid death, and was finally lifted out at fifty-six degrees hard and stiff as a board. Goltz's own summary of the second animal is the origin of the whole trope: the brainless frog lets itself be slowly boiled without so much as moving its legs apart.

The escape term belongs to the apparatus and not to the animal, which the same paper establishes and the received telling drops. In a second run Goltz used a frog whose brain was cut across in front of the optic thalami; it sat still to about thirty-two and a half degrees and then, at thirty-five, jumped clean out of the vessel — and out again on each of six replacements, dying in convulsions at forty-two only because it was put back each time. So a brain-bearing

frog that can leave, leaves. Sedgwick (1888), whose summary is what most later accounts quote, records the same thing with the condition attached: the normal frog leaps away, *or, if confined,* becomes violent in his attempts to escape.

So the two preparations are the two failure modes this section has separated, and the confinement is a third term the popular version discards. Perception removed — checks bypassed, measures withdrawn, the account no longer held — is the second frog. Perception intact, escalation full, and no exit available is the first, and it is the one this section mostly documents: an engineer who warned at the right threshold, a court that determined correctly, publicity at its maximum. *Struggling is not the same as being able to get out*, which is the whole of the adaptability paper's Remark 11.3 in a sentence — and Goltz's vessel is the closed exit made physical.

The popular telling describes neither preparation. It has an animal that does not notice and therefore does not struggle — the brainless one — while purporting to describe the animal with a brain, and so attributes to inattention what the experiment attributes to a vessel with smooth walls and a hand that puts the animal back.

And the prescriptive error follows from the descriptive one, which is why this is worth more than a correction. A failure located in perception prescribes awareness: more warning, better monitoring, wider publicity. Remark 6.3 measured that conversion at approximately zero. So the received story recommends precisely the intervention this section found inert. A wrong model of a failure mode aims the remedy at the wrong term, and that is the general statement, of which the frog is merely the memorable instance.

8 Counterparties, and the price of a promise

Proposition 8.1 (Sustaining is a threshold, so a suppressed decline arrives discontinuously)**.** *An arrangement held up from outside has its residual absorbed by the supporting party, so the signal the decline would otherwise produce is suppressed rather than absent. Support continues while it costs the sustainer less than the alternative and stops when it does not, and a comparison of two costs flips discretely. A continuous decline read through a threshold-valued decision therefore produces a discontinuous observable: apparent suddenness is proportional to how well the signal was suppressed, and the timing belongs to the sustainer rather than to the declining party.*

Remark 8.2 (A propped arrangement buys the absence of its own signal)*.* Two corrections to the obvious reading, the second of which explains why a foreseen transition still arrives as a shock.

Recovery is possible and bounded, and the bound is already stated. Concentrating authority does restore reachable deformation at a centre, and nothing above denies it. It restores it only in the frames authority reaches. By the adaptability paper's Remark 13.4 there is no conversion into individual capacity at any price, the rate into expertise is a career, and the rate into materiel is a production line — so recovery at the top does not reach the buffers whose depletion this section has been tracking, and a centre may become more able to act while the arrangement becomes less able to do anything.

And an arrangement held up from outside has its residual suppressed, which is purchased rather than natural. A party with an interest in continuity absorbs deformation that would otherwise appear as signal. So the quiet is bought. Remark 7.2 found no residual and therefore no adaptation; here there is residual and someone is paying to keep it from surfacing, which looks identical from inside and is not the same condition at all.

Sustaining is a threshold and not a gradient, which supplies the mechanism the previous remarks lacked. Support continues while it costs less than the alternative and stops when it does not, and a comparison of two costs flips discretely. So a continuous decline, read through a threshold-valued decision, produces a discontinuous observable. Nothing in the world jumps. The response jumps, and the response is what is seen.

Which inverts the comfortable reading. Apparent suddenness is proportional to how well the signal was suppressed, so stability maintained by external support predicts a fast transition rather than a slow one. And the timing belongs to the sustainer: support ends when the sustainer's balance turns, not the declining party's, so a declining arrangement does not control the timing of its own phase change.

And a split conducted in the open is suppression failing rather than holding — a residual surfacing where it had been contained. But it is that arrangement's residual, in that arrangement's frame. Nothing licenses netting it against the capacity of another: *Narrativium*'s Proposition 8.1 forbids the cross-frame sum, and reading an elite split in one state as evidence about institutional capacity in another would be precisely the error this paper spends its length refusing.

Stated generally and about nobody. Which parties are sustaining which arrangements is inference from actions, and by the adaptability paper's Remark 7.2 consistency with a model is not identification of one. The result holds of any propped arrangement, including anything this programme builds, and it supplies no date.

Remark 8.3 (A fixed term is a published boundary condition, and can be waited out). One party has declined to resume negotiations with another after the memorandum noted above expired without agreement (Al Jazeera, 2026b): no decision to resume had been taken by mid-August, messages passing through intermediaries being expressly not called negotiations (Iran International, 2026), and at the beginning of September talks were still not being held, each side saying it did not want them on the other's terms (Schneid, 2026). The rationale has been attributed variously — in a longer form as a wait until the counterparty's fixed term ends on 20 January 2029 (FXStreet, 2026), and in a narrower one as no resumption while the terms agreed in June are being violated (Iran International, 2026). Which of them was said, and by whom, does not matter, and dwelling on the attribution would be careful about the wrong object. The behaviour is the datum: talks available are not being held.

Declining to transact when transaction is on offer is a revealed preference, and the calculation is recoverable from the choice without anybody's assistance in stating it. Later is being valued above now. Two things would make that correct, and neither requires a spokesman: that the counterparty's capacity continues to fall, so what is obtainable later is greater or cheaper; and that a later counterparty might be able to deliver what it agrees to.

Declining to negotiate withholds a conversion. Negotiation is the instantiated relation by which pressure becomes settlement, and refusing it produces no deformation now, deliberately. That is a choice about the κ of *Narrativium*'s Definition 8.2 — a rate chosen rather than a rate absent, in the sense of the adaptability paper's Remark 13.4.

Announcing the refusal is Remark 5.2's operation aimed at an entire negotiating position. Declaring a counterparty unable to conclude anything changes what every *other* actor represents as reachable through them — so it devalues every negotiation that counterparty is running, globally, at no cost. *The cheapest attack on a negotiating position is not defeating it but stating publicly that it is not worth transacting with.*

And a fixed term is the most reliable published boundary condition available anywhere. A Seldon manifold (*Seldon*'s Definition 6.1) is built from represented future boundary conditions, and those ordinarily have to be inferred from behaviour, which the adaptability paper's Remark 7.2 says underdetermines them. A term limit is constitutional and dated, so waiting stops being hopeful and becomes computable: the wait has a length, and everyone can read it.

The reflexive cost is general and belongs to no particular party. Any time-limited actor has supplied that input to every counterparty and cannot withdraw it, so the asymmetry between term-limited and unlimited actors is published rather than inferred, and it is structural rather than a matter of skill.

Which suggests the refusal is a solvency assessment rather than a snub, and that is the better reading. A commitment is a promise of future deformation, so an agreement is worth what the counterparty's capacity to perform is worth. Where narrative exceeds Narrativium the promise

is not backed — and by Remark 4.5 the deliberative channels that would make a commitment binding are the ones carrying least. An actor in that condition cannot be contracted with in good faith or bad, since the obstacle is not intent.

It also unifies the two versions rather than leaving them apart. An agreement signed and then violated is direct evidence of inability to perform, so the conditional refusal and the term-length one are a single assessment held at different confidence.

And it names a consequence of the gap this section has been measuring that was not yet stated. An arrangement whose narrative outruns its Narrativium does not merely fail internally. It ceases to be transactable, progressively and independently of anyone's intentions, because counterparties price commitments by capacity to perform and that capacity is what has gone. Losing the ability to make agreements is a cost of the gap, not a separate misfortune.

Waiting is not free either, which keeps this from being a dominant strategy. The waiting party depletes across the same interval, so the position is a bet that the other depletes faster — which is the comparison of the adaptability paper's Remark 13.1 run between two parties instead of within one.

Remark 8.4 (The promise is quoted continuously, and the quote is not narrative). The preceding argument is not an analogy for a bond. A bond is the object. It is a promise of future deformation, priced by counterparties with money at risk, and the price is published every day.

Yield is what the promise fetches; term premium is the price of doubt about capacity further out, which is precisely the quantity Remark 8.3 says a counterparty assesses. In August 2026 a thirty-year auction cleared at 5.216 per cent, the most paid for such debt in a quarter of a century, with bid-to-cover at 2.39 and primary dealers taking 11.5 per cent, both below their twelve-month averages, and the yield awarded above the when-issued level — demand weaker than expected, on a measure published at the time (CRFB, 2026b; Ritchie, 2026). The decomposition is what matters: a move carried by term premium rather than by expectations of policy is the one that speaks to capacity, since a policy-driven rise would say nothing about it. Term-premium estimates are model-dependent and contested, and that is a real qualification rather than a formality.

Which makes this the one measure in the section that is not narrative. Every other assessment canvassed above is somebody talking. This is people paying to hold a view, continuously, in public, and being wrong about it costs them. It is Narrativium, quoted.

The two clocks appear directly in it. Short promises price close to unremarkably and long ones price worse, which is near-term capacity accepted and far-term capacity doubted — the two-clock structure of the adaptability paper's Remark 12.9 read off a curve rather than argued for.

And the nominal-against-real distinction is the narrative-against-Narrativium distinction exactly. A sovereign borrowing in its own currency always keeps the nominal promise, because keeping it costs nothing — that is the narrative, and it is never broken. Whether it keeps the real promise is capacity, and the market prices the two separately, so the gap between them is quoted rather than inferred.

Two facts are kept apart that are easily merged. Foreign holdings fell over the period, and indirect bidding at that particular auction was strong. Those are different measurements of different things, and one auction is not a trend.

9 What this rests on, and where to attack it

Remark 9.1 (Nothing above requires believing anyone's account of their own conduct). The section rests on no testimony, which is worth auditing rather than asserting, since it is the property that decides how it can be attacked.

Every load-bearing input is one of five kinds. Published measurements: projections, debt stocks, price-to-income series, interceptor counts, staffing counts, democracy indices. The

existence of documents, as distinct from anyone's account of them. Procedural records: what passed, through which channel, on what margin, and what a filing contained. Contract terms: whether a tender was competitive, its amount, its awardee. Prices: what an auction cleared at. *And observed acts*: talks not held, a deployment extended, charges dropped. Not one of them is a person's account of what they did.

Where a self-account does appear it is used as an event and not as evidence. That an explanation changed is a fact about the record; neither version of it is relied upon. The published objectives are treated as a document that exists, and Remark 5.3 explicitly declines to attribute any particular act to them. The rationale for a refusal was demoted precisely because the behaviour, not the reason given for it, carries the weight.

And this is forced by the account's own results rather than chosen out of fastidiousness, on three independent grounds.

The first is strategic and is the familiar one. By Remark 5.2 a public statement is an intervention on manifolds rather than a report of one, so an account of one's own conduct is an act aimed at whoever receives it.

The second is not strategic at all, and it is the one usually missed. A self-model is a model and can be wrong. The fourth cost of Graphic Equalisation's Definition 7.13 is exactly this: a graph that has spent without effect acquires a false model of itself, in which something was changed that was not. An account of oneself is read off that model, so its accuracy is the coherence between the model and what it models, which this programme treats as a variable and not a constant. So sincerity is no repair. The usual answer to "they may be lying" is to establish good faith, and good faith leaves this untouched.

The third is constitutive and makes the preceding paragraph's phrasing too weak. Communication is the deformation of a receiver's manifold — that is not a side effect of communicating but its success condition, since an utterance that changes nothing held has communicated nothing. So there is no unaimed variety. Testimony is not a corrupted instance of an otherwise clean channel; there is no clean channel, and that includes honest speech, and it includes this paper, which is aimed at its reader and says so.

Which improves the criterion rather than merely defending it. The axis is not honest against dishonest but *aimed against unaimed*. A price, a count, a filing, a contract term and an act exist for reasons other than reaching you, and can therefore be *observed* rather than received. *The rule generalises past this section: prefer inputs produced for reasons other than your reading of them*, and treat everything produced in order to be read as an intervention first and a description second, whatever its author intended.

The consequence for attack is narrow and public. An argument resting on testimony falls to a recantation or a rival witness. This one falls only if the counts, filings, prices and procedural records are wrong, and all of them are checkable by anyone.

The same property is the limit. Intent cannot be established this way and is nowhere claimed, because the section holds no instrument capable of it.

Remark 9.2 (The author's preferences, and why they are not load-bearing). An author holding views about the subject may wish to state which way they run, and the temptation is to offer the direction as reassurance — that the conclusion was reached against the preference rather than toward it. The preceding remark forbids accepting that, and it forbids it here as much as anywhere: a paper cannot admit its own author's account of himself while refusing everyone else's, and the three grounds apply to him unchanged, the second most of all.

It does not need to be accepted, which is the actual reassurance. The section was built to be independent of testimony, and the author's is testimony. A conclusion that would change if the author's preferences were different would be a conclusion resting on the author; this one does not, and that is checkable rather than promised — the auditability condition of the adaptability paper's Remark 10.1 applied to motive instead of to posits.

But the genuine exposure is selection, and the kind of an input does nothing about it. A

false picture can be assembled entirely out of true prices, true counts and true filings by choosing which ones to assemble. That is where bias actually enters, and no amount of care about admissible kinds excludes it.

And this section is exposed there specifically, which is recorded because it is so. The cases examined above arrived as prompts and were followed up; no protocol enumerated the space of comparable cases and sampled it, and no systematic search for counter-instances was run. A reader's strongest attack is therefore not that the author wanted a conclusion but that these were the cases looked at.

Which means *undeniable* overstates it. By Remark 11.2 the shape is deniable by attacking any of three structural steps, and it is deniable by attacking the selection just admitted. *What it is not is deniable by attacking the author* — a narrower claim, and the only one a declaration of interest can actually secure.

Remark 9.3 (The opposition, put as strongly as it can be put)*.* One class of objection can be dismissed and should be named so that it is not mistaken for the rest. A short window read against a long trend is an aperture error: a single quarter's improvement set beside a movement of an order of magnitude over a decade is not a reversal, and arguments of that shape are artefacts of the window chosen. Four objections are not of that kind.

Objection one: the reference class. The first is the reference class, and it is the strongest. Every argument for structural decline ever made has had this evidence structure — real measurements, real institutional failures, a coherent mechanism — and most of them were wrong. The inputs were checkable in those cases too. And it sharpens: if the shape survives error in every magnitude, as Remark 11.2 claims, then what observation would have counted against it? An account robust to all magnitudes may be unfalsifiable in practice, and that charge is not answered here.

Objection two: the instrument sees only depletion. The second is that the instrument sees depletion and is nearly blind to creation. Counting what has left, what has narrowed and what has been drawn down cannot register new capability, substitution or institutions that did not previously exist. The section's own citation cuts this way: competence migrating into readable form is, by the adaptability paper's Remark 9.2, the largest capability gain the argument contains — paid once, copied freely — and it is read here only as a risk.

Objection three: the stock against the flow. The third is the stock against the flow, and it uses this paper's own rule. The dollar held about 57.1 per cent of official foreign exchange reserves in the first quarter of 2026, with depth and liquidity that are market structure rather than narrative (IMF COFER, 2026). Every counterparty, aggregated, with money at risk is by Remark 9.1's own criterion better evidence than any single actor declining to transact.

Objection four: cyclical rather than terminal. The fourth is that the mechanism may be cyclical rather than terminal. Comparable or worse institutional dysfunction has occurred repeatedly and been recovered from, and nothing above distinguishes an oscillation from a trend.

Reply to the second: a gain is an instance of the phase change, not a counterweight. The second objection fails, and it fails on this account's own machinery rather than by rebuttal. The adaptability paper's Remark 12.1 defines a phase change as a change in the dimension set — which is direction-neutral. Additions are changes in it exactly as removals are, so a capability gain is not a counterweight to the phase change but an instance of it. That remark says as much already: new dimensions *add* distinguishable reachable states, so a projection over fixed dimensions understates the change. Gains therefore make the projected change larger, and the objection had the sign inverted.

Change is risk without regard to direction, which is the priors argument applied to novelty. A held set is matched to the frame in which it was acquired, and Graphic Equalisation's Definition 7.13 is explicitly relative to the work being done — so a gain de-matches everything holding priors fitted to the world before it, and the de-matching does not care which way the change ran.

And gains and losses cannot be netted, which is where the objection actually goes wrong. The same event — competence becoming copyable — is a gain in the capability frame and a shock to every frame that priced the scarcity, which is Remark 3.1. Opposite signs, one event, and no instantiated conversion between the frames. So summing them is the cross-frame sum *Narrativium*'s Proposition 8.1 forbids, and the objection smuggled one in — the same error caught three times already above.

What keeps this from confirming everything is a rate condition, and it is observable. A gain propagating at copy speed into frames that absorb at acquisition speed widens the distance the adaptability paper's Remark 12.9 measures; a gain absorbed at the frame's own speed does not. The comparison of the two rates decides it, not the sign of the change — so not every capability gain counts for this argument, and which ones do is a question with an answer.

Reply to the first: the class was selected by discomfort, and memory is the wrong instrument. The first objection is stated at the wrong grain. Its reference class was civilisational decline predictions. Three things are wrong with that class.

The base rate of the event is not low; over long horizons it approaches one. Every arrangement that has existed, excepting those extant, has changed phase. What has the poor record is *timing*, and this account declines timing explicitly — the adaptability paper's Remarks 12.1 and 12.9 give a reachability difference and two clocks and no date. So the failed predictions attack a claim not made here.

The grain is wrong in a second sense: the mechanism is scale-free. Frames, channels ordered by friction, and conversions between them apply to a firm, a department, a ship. At those grains instances are plentiful and outcomes are checkable, so the mechanism can be tested where evidence is abundant instead of only where it is contested and singular.

And structural failure prediction is ordinary, professional and calibrated. Credit assessment, actuarial practice and reliability engineering forecast structural failure daily and are scored on it. The claim that such prediction does not work is false as stated; what is true is narrower, and concerns singular events at civilisational scale over long horizons.

What survives is the unfalsifiability charge, which the objection conflated with the base rate and which is a different complaint. It is now partly answered: a rate condition distinguishes which capability gains count, and a curve shape distinguishes preference from lock-in. A retrofitted falsifier is worth less than a pre-registered one, which this programme states elsewhere and does not get to forget here; both conditions are accordingly entered as commitments in Section 12, where they can be checked against what follows rather than against what preceded them.

There is a fourth thing wrong with that class, and it is this account's own mechanism turned on the objection. The class was populated by memory, and memory is residual-driven: by Remark 7.2 what is encoded is what surprised. A prediction that came true and was absorbed generated no residual and left no trace, which is why structural forecasting that works — credit, actuarial, engineering — is recounted by nobody. *The remembered sample is selected for having been wrong, because being wrong is what made it memorable.*

And a correct warning that is acted upon erases its own evidence, which is the sharper half. Such a prediction is vindicated by the absence of the thing predicted — and absence is exactly what a false alarm also produces. The observational signature of an averted disaster is identical to that of a groundless one, so successful prediction is systematically misfiled as failed prediction, and the better the response the more complete the misfiling.

So the base rate cannot be read off memory at all. The claim is not that it is higher than

supposed but that memory is the wrong instrument, biased in a direction that can be derived rather than merely suspected, by two mechanisms working the same way.

Which collapses two residuals into one. What the falsifiability charge needs is a falsifier declared before the fact; what the base rate needs is a record of predictions scored after it. Both are pre-registration, so the promissory note above is a single note rather than two.

And it applies reflexively, uncomfortably. If this account is right and anything is done in consequence, it will be indistinguishable from having been wrong — by the mechanism just given. The only defence is to register the commitments before they resolve, which is what the falsifiability section exists for and is the reason it is not decorative.

And underneath the reference class was something plainer, which is worth naming because it is the default position rather than an eccentric one. The class was selected by discomfort. A set defined by alarming content was chosen, its poor record observed, and the fact that the alarm was the selection criterion left unnoticed. The operative rule is the more uncomfortable a prediction, the more likely it is to be wrong, and it wears the costume of calibration.

It feels like calibration because it is a well-fitted prior applied outside its frame. Where most alarming claims are encountered — advertising, solicitation, panic — alarm and exaggeration genuinely do correlate, so the heuristic earns its keep there. Graphic Equalisation's Definition 7.13 is relative to the work being done, and this prior is well-formed for that work and not for structural measurement.

Worse, it filters on the axis the two cases share. Alarming and wrong is cheap and common; alarming and right is rare and consequential; and the filter selects on alarming, discarding the second along with the first. It is not a weak instrument but one aimed at the wrong feature.

The deep form is that discomfort is a residual. An unwelcome prediction is one that does not fit what is held. So treating discomfort as evidence against the claim is treating a residual as disconfirmation rather than as a trigger to update — the adaptation mechanism run backwards.

Which adds a failure mode to Remark 7.2 rather than restating it. That remark has no signal generated and therefore no adaptation. This has signal generated and discarded because it is signal, and the two are independent.

And it is a third preparation the frog experiment does not contain. Goltz has perception absent in one animal and exit absent for the other. The human case adds noticing, and concluding that the water is not hot because the conclusion is unwelcome — an update failure sitting between perception and exit, available only to a system whose self-model has a stake in the answer.

The voicing filter, which sits upstream of all four. And the objection's form is social rather than epistemic, which three marks establish. It is *content-independent*: the charge of alarmism applies identically to a true alarming claim and a false one, whereas an epistemic objection engages a claim and this one engages a category. The cost falls on the speaker rather than the claim, so the sanction attaches to the act of voicing and not to the proposition, which is the signature of a norm and not of an argument. And it is asymmetric: there is no comparable penalty for being wrongly reassuring, so the expected standing of an alarming prediction is negative *whatever its accuracy.* The rule need not be believed by anyone. It need only be enforced.

Which compounds the selection already described. One filter operates downstream, in memory, keeping the alarming predictions that failed. This one operates upstream, at the point of voicing, before any assessment is possible — so claims are removed from circulation before they can be checked, and the surviving public record is selected against such predictions being right *twice.*

In this account's terms it is Remark 5.2's operation aimed at a speaker. A reliable expectation of cost changes what the speaker represents as reachable, and it is the cheapest intervention in the catalogue: no institution, no enforcement body, no decision. And it is Remark 4.5's ordering

once more, since nobody resolves to suppress anything — the speaker simply does not say it, and change again accumulates in the channel where nothing is decided, here determining what enters the record at all.

So the path from an accurate model to no action has three stages, and each is sufficient alone. A decline arriving as predicted generates no residual. A residual that is generated may be discarded as discomfort. And a residual neither absent nor discarded may simply never be uttered, because uttering it is priced. The account has now documented all three separately, and they compose: holding the right model, recognising it, and saying nothing are jointly consistent, and jointly sufficient for nothing to happen.

And the small-scale test is named rather than run. Nothing above examines a firm, a department or a single institution at the grain where the mechanism could be checked cheaply and often. So the answer to the first objection is at present a promissory note, and it is recorded as one.

Reply to the third: read the flow for preference and the stock for the cost of exit. The third objection inverts as well. A reserve holding is a conversion outward by the holder: a frame turning its own reachable deformation into a claim on another's. So the fifty-seven per cent is not a measurement of the dollar's capacity. It is a record of what other frames have done with theirs — Remark 8.2's propping, itemised, with the props bought and paid for by the propping parties. The objection reads a stock as a property of the thing held rather than of the holders' allocation, which is the recurring error of this section's subject matter and not a peculiarity of the objection.

That reading must be falsifiable, since the first objection still stands, so here is what separates it. If holding reflects preference, the share should be stable or rising over long horizons. If it reflects lock-in, the share should decline slowly and persistently while fast exit remains self-harming, since selling destroys the value of what is still held. The two predict different curves.

The record shows the second. From above seventy per cent in the late 1990s to 57.1 per cent now: slow, one-directional, and unaccompanied by any holder moving quickly (IMF COFER, 2026). The recent quarterly rise is small and about half valuation, which is not a reversal — and on this reading it is not even a counter-instance, being a price effect on a position nobody can exit at speed.

Which places the holders in the basin of the adaptability paper's Remark 11.3 rather than the issuer alone. Each year's continued holding is an accommodation; the accumulated position raises the cost of leaving; and what presents as confidence is a measurement of exit cost. It also sharpens Remark 8.2: a sustainer sustains not from choice but because stopping is self-harming, and the flip arrives when continuing costs more than the harm of leaving.

The rule that falls out is short. Read the flow for preference and the stock for the cost of exit. At end-2025 market prices gold was 27 per cent of total official foreign reserves against 22 per cent for United States Treasuries, having passed them for the first time; corrected for valuation at the end-2023 gold price the shares are 16 and 26 per cent, so the crossing is a price effect and not an executed reallocation (ECB, 2026b). Read as a flow it still points toward an asset that is nobody's liability; read as a stock it says nothing yet.

None of which repairs the exposure. The reserve position enters this section only because the objection was sought deliberately, which is Remark 9.2's selection exposure found operating rather than merely conceded.

The fourth objection is not answered. Nothing above distinguishes an oscillation from a trend, and no instrument here could: the account supplies shapes and declines dates, so it has no way to say whether a phase change already under way is one an arrangement has been through before. That is a limit and is recorded as one rather than argued away.

10 One seat, examined to the bottom

The single seat below is an instrument, not an accumulation of grievance: it is examined to establish one result, that institutional transitions are keyed on records rather than on states (Proposition 10.9). Holding that payoff in view is what the remarks that follow are for.

Remark 10.1 (The voicing filter has a price, and releasing it is observable). The preceding filter is priced, which means it can be measured by watching what appears when the price falls. One case supplies a within-person control.

A long-serving legislator announced on 28 February 2024 that he would give up the party leadership he had held for seventeen years, and left it when his successor took the post on 3 January 2025 (McIntire and Lesniewski, 2024). Within weeks he was among the very few of his party to vote against administration nominees, doing so on 24 January, 12 February and 13 February 2025 and stating specific grounds (Wilkinson, 2025). What changed was the price of dissent, not the information available. An account on which such objections are epistemic predicts no change from that; *behaviour tracking price rather than evidence is what a filter predicts*, and it is what occurred.

The retirement announcement of 20 February 2025 (Davis and Schapitl, 2025) is not used, and reporting why is worth more than the datum would have been. It post-dates all three votes, so it cannot be their cause. That is exactly the announcement-date-against-behaviour check this remark's own study design prescribes below, applied to its own instance — and applied here it fails one of the two candidate causes, leaving the leadership exit as the price fall that precedes the behaviour and discarding the other.

What is not claimed is that he held the view earlier and withheld it. That is a mental state, and Remark 9.1 excludes it whoever offers it. Only the coincidence is observable: a change in expressed position arriving with a change in what expressing it costs.

And this supplies the study owed above. Exogenous falls in the cost of dissent — a leadership post given up, a term ending, a departure from office — are dated, numerous and public, and voting records are public too. So the filter's magnitude is measurable across an entire legislature, at the small grain where instances are plentiful, which is precisely the test named and not run. The obvious confound is that retirement is not fully exogenous, since a position may become untenable or a departure may be chosen in order to speak; comparing announcement dates against the timing of behaviour change is what addresses it, and the paragraph above shows that check discarding a candidate rather than merely being named.

The magnitude in this instance is small, which is informative rather than awkward. Releasing the filter produced marginal dissent and not reversal.

And there is a corollary worth stating because it is unpleasant and follows directly. The cost of dissent falls as remaining tenure falls, and so does influence. The two decline together, so dissent becomes affordable exactly as it becomes ineffective, and a late objection is by Remark 4.5 lodged in the channel that carries least. A filter released only at the exit is a filter that was never really released.

Remark 10.2 (Competence accumulates over a lifetime, so it concentrates where it cannot be copied). The same case carries a second reading, and it concerns nobody in particular. Critical roles were held through episodes of illness, and continuity depended on those episodes not being terminal — a four-month absence later disclosed as depression, a member moved to assisted living with dementia and not disclosed at all, and the historical practice of justices consulting physicians on the timing of retirement (Gringlas, 2026).

A person's held set is not addressable, by the adaptability paper's Remark 9.2. So decades of accumulated procedure, relationship and judgement in a critical role is real capacity with no copy operation available for it, and the arrangement's continuity is contingent on a fact nobody chose and nobody manages.

Nor is there a transition mechanism. Departure occurs by death, by incapacity, or by choosing

to leave: none of the three is scheduled and only the last is chosen. So the timing of a capacity loss in a critical role is set by biology, which is Remark 8.2's finding that the timing belongs to the sustainer, where here the sustainer is a body.

It also gives the crossing of Remark 10.1 at the individual grain, and generally. The events that lower the price of speaking — a leadership post given up, an illness, a short remaining horizon — are the same events that lower the capacity to act on what is then said. The voice is freed and weakened by one occurrence, which is why the release is smaller than it looks and why nothing about it should be read as reluctance.

And the concentration is not a defect, which is the part most easily got wrong. Competence accumulates over a lifetime because acquisition is per-lifetime, so it is *necessarily* densest in those with least of one remaining. *The most competent are therefore the least copyable*, not by any failing but because competence takes a lifetime and lifetimes end. The defect is the absent transfer operation, not the demographics, and treating it as the latter mistakes a missing mechanism for a bad cohort.

Which is this whole section at the scale of one person. The narrative names an institution. The Narrativium is held, in part, inside individuals whose held sets have no copy path — and an institution in that position is one actuarial event away from discovering which parts of itself were never written down.

Remark 10.3 (A loss need not be terminal to be total, and a partial one registers nowhere). Succession is not where the difficulty lies. Succession asks what becomes of a held set when it is finally lost; the record shows the loss already occurring, partially, for weeks, with nothing registering it.

A member was hospitalised from mid-June following a fall, moved afterwards to rehabilitation, and missed dozens of floor votes; no public account of the condition was given for several weeks, and a longstanding mobility condition was afterwards described as a continuing factor (NPR, 2026).

The stated composition and the available capacity diverged, and no published measure of the chamber distinguishes them. A roll call records an absence, and an absence is the same category whether its cause is a hospital or a fundraiser. So the institution held a false model of itself — a seat counted, committee positions held, a member present in every enumeration — while the capacity was not there. That is the fourth cost of Graphic Equalisation's Definition 7.13 exactly, and it was silent exactly as that definition says it is.

And no mechanism registered it because there is none. Nothing declares a temporary incapacity, seats a substitute, or compels disclosure. So the change occurred in the channel where nothing is decided, which is Remark 4.5 once more, and here for the plainest of reasons: there was no other channel available.

The only instrument was testimony, released at the subject's discretion. At the one point where an institution most needs a capacity measurement of its own members, it holds nothing but the input class this account ranks least reliable, on three grounds, delivered on the subject's timetable. That is not a criticism of anyone's conduct. It is a description of an instrument that does not exist.

Nor was any of it surprising, which closes the loop. A longstanding condition is a known risk, so the event generated no residual, and by Remark 7.2 nothing adapted — no contingency before, no mechanism during, and afterwards a statement.

Succession is the comfortable version of this question. The uncomfortable one is that capacity can be absent for weeks inside a fully-staffed institution and appear in no measurement it keeps of itself.

Remark 10.4 (Authenticating an aimed artefact leaves it aimed). The instrument question sharpens further, and the sharpening is methodological rather than medical.

For roughly a month there was no independent observation of the member. What stood in its place was imagery released by his own office, the first of it showing him holding that day's

newspaper, a second following weeks later. Doubt was answered by authenticating the images — media-forensics specialists examined them and found no sign of manipulation or synthesis (McCullough, 2026b,a).

And that did not close the gap, because the gap was never authenticity. By Remark 9.1 an artefact released by an office is *aimed*: produced in order to be received. *Authenticating an aimed artefact leaves it aimed.* The finding establishes that an image was not manipulated and establishes nothing about what an artefact constructed to be received was constructed to convey. The verification performed was orthogonal to the question it was taken to settle, which is worth stating because it was performed carefully and answered honestly.

There is an unaimed signal available, and it is in the necessity rather than the content. A proof-of-life format is produced only where ordinary observation has already failed. The office chose to release; it did not choose that releasing was the remaining option. So the content is aimed and the necessity is structural — read the existence of the artefact, not the picture — and that is a genuine refinement of the criterion above, since the fact that a communication had to be made is frequently unaimed even where the communication is not.

And no institutional instrument stood behind any of it. Disclosure of a member's health in that chamber is discretionary and required by nothing, so what is published is what an office elects to publish (Gringlas, 2026). Nothing in the arrangement produces a measure of capacity to serve.

And a second member of the same body died suddenly in the same weeks, at seventy-one, of a cardiovascular cause recorded by a medical examiner (Ballotpedia, 2026). Remark 10.2 projected a concentration; this is two events in one summer, and no linkage between them is asserted or needed.

Remark 10.5 (The reaction is the measurement, and the test searched the wrong space)*.* The object-level facts are not the finding. Over eighty days have passed, as of 2 September, since a sitting legislator was last independently observed, with office-released imagery in the interval. This is not a scandal. It was a news cycle. *The absence of alarm is the measurement, not the absence of the person.*

And what it measures is that the divergence has already been internalised. Nobody is surprised, which by Remark 7.2 means the observation fits the model held — so the public model *already contains* the proposition that a body's stated composition and its actual capacity are different things. The narrative-against-Narrativium gap is not a finding about the institution. It is a shared prior.

Which is Remark 7.3's result at its strongest. The managing frame held the correct account inertly; the observers held it inertly; and here the general public holds it inertly too, correct at every level and coupled to action at none.

The forensic result is also weaker than it was read to be, and for a reason past detector power. A negative finding on digital manipulation admits three readings, not one. The image was not manipulated. Or it was manipulated well, since such a test confirms presence and never absence, and is asymmetric in exactly that direction. Or — and this is the cheapest by far — the photograph is entirely authentic and the scene was arranged, which requires no manipulation whatever and leaves nothing to detect.

So the test searched the space of forgeries, and an aimed artefact's ordinary method is not forgery. Digital alteration is the hardest route, the only one leaving traces, and the least necessary; reporting it clean rules out the option nobody needed. The analysis was competent and honestly reported, and the inference drawn from it does not follow — which sharpens why authenticity was orthogonal: the test's hypothesis space excluded the likeliest case by construction.

Proposition 10.6 (Aimed artefacts bound; unaimed records estimate)**.** *An artefact produced in order to be received is* selected *by whoever released it, so it reports the best available instance of what it depicts. It therefore bounds a distribution from above and says nothing about its centre:*

it establishes at least once, this*, and cannot establish* typically, that*. A record produced for reasons other than being read is not selected on the reading of it and estimates the centre. Where the two conflict the aimed artefact yields, by the criterion Remark 9.1 gives.*

Remark 10.7 (Selection alone accounts for the tension, and no bad faith is required)*.* Proposition 10.6 follows from selection and needs nothing else — no forensics, and nothing inside the artefact. The check that pays is between the artefact and the record: the image depicts a condition markedly better than an absence of over eighty days, dozens of missed votes and a discharge to continuing rehabilitation would imply. The two cannot both be read at face value, which is a disjunction and is offered as one.

And the ordinary resolution requires nothing exotic, which is what makes it the right one. An aimed artefact is *selected*. A photograph is one instant out of many, chosen by whoever released it, and the worst instant would not have been released. Selection alone accounts for the tension, with no manipulation, no arrangement and no bad faith required.

So the artefact is structurally incapable of the work it was released to do. The question asked for a central estimate; a selected sample cannot supply one. That is a fact about selection and not an accusation, and it holds however the artefact was produced — and reading an upper bound as a summary is the error, one made by everybody who treated the release as answering the question.

Here the unaimed record is a vote count, which nobody released in order to be read, and which by that fact is worth more than the photograph it contradicts.

Remark 10.8 (An open case, which makes the prediction a pre-registration)*.* The episode has not closed. Discharge from rehabilitation was reported in early August, in a statement from the member himself; the last reported independent sighting is of 11 June, so the count had passed eighty days by 2 September and no confirmed sighting has been reported since (Al Jazeera, 2026). New information has continued to arrive as office statements and third-party accounts of private contact.

Which is worth more to this account than any of the closed cases, and for a reason the section owes. Remarks 11.2 and the discussion of falsifiability concede that the discriminating conditions above were produced under pressure rather than declared in advance, and that a retrofitted falsifier is worth less than a registered one. *An open case admits of registration*, so four predictions are set down here to be checked later rather than argued now.

First, no institutional mechanism will be triggered by the continuing absence: no determination of incapacity, no substitute seated, no disclosure compelled. *Second*, further information will arrive as aimed artefacts — statements, imagery, reported private contact — and not as independent observation. *Third*, the absence will end by voluntary return, by resignation, or by death, and not by any act of the body. *Fourth*, no measure the chamber publishes will have recorded the gap: the record will show absences of a kind indistinguishable from any other.

The weakness of this is worth stating, since predicting inaction is cheap. Inaction is the default, so the first and third carry little on their own. The content is in the second and fourth, which say what kind of evidence will appear and what the record will look like afterwards, and neither is a default.

One methodological note, because it bears on the reliability of everything above. A search on a live case returns material from earlier episodes involving the same person, and a careless reading merges them into the present one. Only material dated to this episode is used here. A live record is the condition under which that error is easiest to make and hardest to notice.

Proposition 10.9 (Institutional transitions are keyed on documents, not on states)**.** *Consequences inside an institution are triggered by records and not by states of the world. A certificate initiates succession; the death does not. A determination seats a substitute; the incapacity does not. An institution's reachable set is therefore a function of what has been recorded rather than of what is the case, and whoever controls the record controls the transition independently of*

the underlying fact. The divergence between an institution's stated composition and its actual capacity is accordingly invisible to its own machinery in principle, and not by neglect.

Remark 10.10 (The world enters only through documents, so the gap is invisible in principle)*.* The right question is what manifold withholding alters, and it can be answered without settling anything about the case that prompted it.

Proposition 10.9 follows from the adaptability paper's Remark 11.3 exactly: the cheap exit is revoking a frame, revocation here requires a particular filing, and absent that filing the frame remains instantiated whatever the world does. The zero-cost operation is being held *shut* rather than taken, and holding it shut requires nobody to assert anything at all.

And it generalises past this instance, which is what makes it a result. Insolvency arrives by filing, not by being insolvent. Incapacity arrives by determination. Vacancy arrives by declaration. *An institution's model of the world is its document set*, and the world enters only through documents.

Which completes the section rather than adding to it. That no measure the chamber keeps records the gap between its stated composition and its capacity is *not an oversight.* It is constitutive. An institution reading actual capacity would be reading the world directly, and none of them do — they read filings, so the divergence this section has measured throughout is invisible to the machinery in principle and not by neglect.

The surrounding rumour is handled by the rule already given and requires no exception. A widely held belief is aimed communication in aggregate, which this account ranks least reliable, and nothing here asserts anything about the underlying state, nor needs to. What is observable is that a widely held belief has produced no institutional consequence whatever — which is the same finding from the other side. Neither facts nor beliefs are inputs. Filings are.

Corollary 10.11 (A certificate is the conversion relation)**.** *A record of the kind Proposition 10.9 names is not a report about the world but the instantiated conversion of* Narrativium*'s Definition 8.2 between the frame it records and the institutional frame — a death certificate is the κ between the biological frame and the institutional one, which is precisely why it triggers succession while the event it records does not. Withholding it withholds the conversion, so by* Narrativium*'s Proposition 8.1 no accounting crosses, whatever is true on either side. The document is not evidence; it is the bridge — and that is why withholding one is effective without anybody lying.*

Remark 10.12 (Why the specific state is irrelevant, and where the interval has been used)*.* Corollary 10.11 follows from the frame reading, and this account can say why the specific state is irrelevant rather than merely conceding it. To be is to stand in relation, so presence in a frame is edges in that frame. The institutional frame's predicates are votes cast, statements made, meetings attended — and on those the answer has been determinate for over eighty days; the κ between that frame and the biological one is *Narrativium*'s Definition 8.2 instantiated in a document, which is the adaptability paper's Remark 13.4 at the smallest grain it has. Biological state belongs to a different frame entirely, and the question that seems unanswerable is unanswerable only because it is being asked across two of them at once.

Parsimony then selects the reading requiring nobody to lie, and this section's own ordering explains why. By Remark 4.5 the channel demanding no decision carries the most. *Not volunteering requires no decision; asserting a falsehood requires one.* So the mechanism itself favours the silent-and-accurate arrangement over the deceptive one — the charitable reading arrived at structurally rather than out of manners, and the more probable for it.

And the separability is documented rather than hypothesised. A head of state's duties were transferred to a successor during illness in 1974, *reversed upon recovery*, transferred permanently three weeks before death, with the successor proclaimed two days after it (Preston, 1993); a Soviet leadership settled its succession while the leader lay dying, published news of the illness only the day before, and held the announcement of the death for a further six hours (RFE/RL,

2016; Service, 2004). Neither case requires falsification of anything. Both show that the interval between a capacity ending and a record being made is real, separable, and available to be used.

None of which establishes anything about the present instance, and it need not. It establishes that the question is frame-relative and that the certificate is the bridge between the frames — which is exactly why withholding one is effective without anybody lying.

11 What follows

Remark 11.1 (A change in how, and not only in what). The remarks above converge on a consequence larger than any of them, worth stating with its conditions attached rather than without.

Conjoin what is established. Individual capacity is fixed and exceeded, so consequential decisions do not fit inside one person (the adaptability paper's Remark 13.2). The deciding unit is therefore the arrangement rather than the person (its Remark 12.6). That arrangement is built and rebuilt by what it contains (its Remark 12.7). And the person's removal from the loop proceeds by capacity rather than by choice (its Remark 13.1). *So what changes is not the content of outcomes but the mechanism producing them* — a change in *how*, and not only in *what*.

That is the work a period named for human agency does. The term's formal geological standing was declined rather than granted — the proposal for an Anthropocene Epoch as a formal unit of the geologic time scale was rejected on a subcommission vote of four for, twelve against and three abstentions, and the rejection was approved and published as a joint statement in March 2024, which recorded that the term nonetheless remains an invaluable descriptor of human impact (IUGS–ICS, 2024) — and what it actually does is periodise by *which agency decides*, which is the register this argument already occupies and the reason it can be used without borrowing authority it lacks. Nothing in it requires humans to go. A period may end while its inhabitants remain, having never been named for their presence.

And it is not remotely new. Arrangements exceeding any individual's comprehension are old: markets and bureaucracies have decided at scales no participant holds for as long as either has existed, and the observation that no individual possesses what the system computes was made exactly (Hayek, 1945). Super-individual deciding is not the finding, and an account offering it as one would be uninformed rather than bold.

What is new is narrower, and is this paper's own subject. Those older arrangements were neither addressable nor self-modifying: the competence in them could not be read out, copied, or altered except through the people carrying it. The arrangements now in question are addressable — the migration reported in the adaptability paper's Remark 10.3 rather than here — and they build, at a rate the human clock does not set. That is the difference, and it is one of schedule and legibility rather than of whether systems decide.

Two limits, the second of which keeps this honest. Humans do not stop deciding and nothing above says they do: what moved is the locus of *consequential* decision, while individual agency persists undiminished at individual scale. And an epoch's ending is a periodisation, frame-relative like every other measurement here — *under the criterion* "which unit decides" the period closes, and Axiom -1 denies that it closes simpliciter. Anyone choosing a different criterion gets a different answer, and is owed the courtesy of being told which one was used.

What ought to follow is not addressed, for the reason given before. This account prices operations. It does not license them.

Remark 11.2 (What carries the weight, and where to attack). The steps above draw almost wholly on results that are canonical or on observations close to trivial. It is worth saying exactly what follows from that, because the obvious reading — so the conclusion is safe — is the wrong one, and taking it would make the argument unfalsifiable.

The conclusions are shapes rather than magnitudes, and that is the whole source of the robustness. The interval is a difference of reachability sets, not a date. The three errors are signed, not sized. The clocks differ in kind, fixed against compounding, rather than by a measured ratio. The deciding unit changed, which is a structural fact and not a quantity. A shape survives error in its inputs' magnitudes so long as their sign and ordering hold, which is why no single empirical step here has to be exactly right.

The robustness is bought, and the price is precision. None of these conclusions says how fast, how much, or when, and a reader wanting those will not find them — the same exchange this paper records everywhere else, not suspended here because the conclusions happen to be large.

So the steps are not equally load-bearing, and pretending they were would be the dodge. The empirical ones are individually dispensable. The structural ones are not, and there are three: Axiom -1 with the frame-relativity results, without which most of the corrections above collapse; the claim that competence is graph-resident with no remainder, without which the adaptability paper's Remark 10.3 fails and the adaptability paper's Remark 9.2 weakens; and containment under coupling, on which the sharpening of the adaptability paper's Remark 12.6 rests. Those are the points to attack, and an attack landing on any of them does damage where an attack on a citation does not.

The claim is better exercised than asserted. The thinnest supports invoked above are the degradation-under-scarcity result (Mani et al., 2013), least settled of the capacity findings, and the externalisation survey (Zhou et al., 2026), which is recent and singular. Remove both. The capacity bound still stands on Miller (1956) and Cowan (2001); the migration still stands on the architectures themselves, which are public artefacts rather than claims; and every conclusion above is unchanged. That is what it means for the shape to be the load-bearing thing — demonstrated on the two weakest links rather than claimed in general.

12 Falsifiability commitments

Each commitment below is marked [SHAPE] or [DETAIL]. A [DETAIL] failing is a revision: the account survives with that mechanism replaced. A [SHAPE] failing costs *range* rather than a mechanism, because what fails is something the substrate is claimed to require or to permit — so the account does not hold where it claimed to, which is the more serious of the two and is still a boundary rather than an annihilation.

This section exists because Remark 9.3 concedes that a retrofitted falsifier is worth less than a registered one, and because Remark 7.2 entails that if this account is right and anything is done in consequence it will be indistinguishable from having been wrong. Registration before resolution is the only defence available against both, and it is the reason this section is not decorative. The commitments are entered as of 2 September 2026.

1. [SHAPE] *Amplitude runs inversely to deliberation* (Proposition 4.4). Order an arrangement's channels of change by the deliberation each requires. *Fails* if the amplitude of change carried does not fall with that ordering — if a contested, high-deliberation channel is observed carrying more than the channels below it, in an arrangement where both are available and both are used.

2. [SHAPE] *Load goes where it is cheapest to place, not where it is best borne* (Remark 6.1). *Fails* if, where friction and slack diverge, load is observed to find slack rather than low friction — that is, if an arrangement under strain is observed routinely to draw on the buffer most able to bear the draw rather than on the one requiring least deliberation to draw on.

3. [SHAPE] *A system that adapts on residual is blind to predicted decline* (Proposition 7.1). *Fails* if a residual-driven system is shown acting on a foreseen, on-schedule decline without an asserted novelty as the trigger — that is, if the arrival of a correctly predicted outcome,

generating no residual by arriving, is nonetheless observed to produce the adaptation that an asserted surprise produces.

4. [SHAPE] *Aimed artefacts bound; unaimed records estimate* (Proposition 10.6). *Fails* if, in a matched sample, artefacts released in order to be received are shown to estimate a distribution's centre as well as records produced for other reasons do — which would make the selection argument idle and the aimed/unaimed axis of Remark 9.1 a distinction without a difference.

5. [DETAIL] *Lock-in, not preference* (Remark 9.3). If reserve holding reflects preference, the share should be stable or rising over long horizons; if it reflects lock-in, it should decline slowly and one-directionally while fast exit remains self-harming. *Fails* on a stable or rising share over a long horizon, or on a fast exit by any substantial holder.

6. [DETAIL] *The rate condition on capability gains* (Remark 9.3). Only gains propagating into frames faster than those frames absorb them widen the distance the adaptability paper's Remark 12.9 measures. *Fails* if a gain absorbed at the receiving frame's own speed is observed to produce the same signature, since the condition would then discriminate nothing.

7. [DETAIL] *Sustaining is a threshold, so a suppressed decline arrives discontinuously* (Proposition 8.1). *Fails* if an externally supported arrangement's transition is observed to be gradual rather than discontinuous — support tapering with the decline rather than flipping against it.

8. [DETAIL] *Institutional transitions are keyed on documents, not on states* (Proposition 10.9). *Fails* on an institution whose transition is triggered by a state of the world with no record standing between — automatic vacancy on a fixed number of certified absences, say, or a seat emptying by operation of a clock nobody files against — which would make the document-keying a design choice available to be revised rather than constitutive of what an institution can read.

9. [DETAIL] *Publicity converts efficiently into damage and inefficiently into repair* (Remark 6.3). The discriminating study is specified and *has not been run*: the rate at which oversight channels activate per comparable maximally-public event, then against now, counted as referrals made, hearings held, contracts terminated, and suspensions and debarments entered. *Fails* if that rate is found not to have fallen, or to have fallen no faster than the rate at which publicity converts into departures.

10. [DETAIL] *The voicing filter is measurable* (Remark 10.1). The discriminating study is specified and *has not been run*: across an entire legislature, exogenous falls in the cost of dissent — retirement announced, a term ending, a departure from office — set against public voting records, with announcement dates compared to the timing of behaviour change to address the confound that a departure is not fully exogenous. *Fails* if expressed position does not track the price of expressing it. Applied to the single instance in Remark 10.1 that check already discards one of the two candidate price falls, the retirement announcement post-dating the votes it was offered to explain.

11. [DETAIL] *The four registered predictions of Remark 10.8*, entered 2 September 2026 on an open case and to be checked on its close. *First*, no institutional mechanism will be triggered by the continuing absence: no determination of incapacity, no substitute seated, no disclosure compelled. *Second*, further information will arrive as aimed artefacts — statements, imagery, reported private contact — and not as independent observation. *Third*, the absence will end by voluntary return, by resignation, or by death, and not by any act of the body. *Fourth*, no

measure the chamber publishes will have recorded the gap: the record will show absences of a kind indistinguishable from any other. The first and third carry little on their own, inaction being the default; the content is in the second and fourth, which say what kind of evidence will appear and what the record will look like afterwards, and neither is a default.

One commitment this section does not make. No date is registered anywhere above, and none is withheld out of caution: by Remark 11.2 the conclusions here are shapes rather than magnitudes, and a shape has no date to register. A reader wanting one is owed the statement that this account does not supply it and does not claim to.

References

Al Jazeera. US Senator McConnell says he's been discharged from rehabilitation centre. 6 August 2026. Records that he had not been seen in public since 11 June 2026. https://www.aljazeera.com/news/2026/8/6/us-senator-mcconnell-says-hes-been-discharged-from-rehabilitation-centre

Al Jazeera. US–Iran Memorandum of Understanding expires: how and why it fell apart. 17 August 2026. https://www.aljazeera.com/news/2026/8/17/us-iran-memorandum-of-understanding-expires-how-and-why-it-fell-apart

Briana Ryan. Lindsey Graham becomes the fourth member of the U.S. Senate since 2013 to die while in office. *Ballotpedia News*, 13 July 2026. https://news.ballotpedia.org/2026/07/13/lindsey-graham-becomes-the-fourth-member-of-the-u-s-senate-since-2013-to-die-while-in-office/

Alison Bath. Navy carrier in Mideast at sea for more than 200 days, testing crew resilience. *Stars and Stripes*, 9 July 2026. Records the 207th consecutive day at sea without a port call, passing the 206 days set by another carrier in 2020. https://www.stripes.com/theaters/middle_east/2026-07-09/abraham-lincoln-middle-east-sailor-mental-health-22219830.html

Kaylyn Barnhart Batista. Home at last: USS Gerald R. Ford returns after record-setting deployment. *Stars and Stripes*, 16 May 2026. Returned to Naval Station Norfolk after 326 days. https://www.stripes.com/branches/navy/2026-05-16/ford-carrier-strike-group-homecoming-21680687.html

Bank for International Settlements. *OTC foreign exchange turnover in April 2025.* Basel, 2025. https://www.bis.org/statistics/rpfx25_fx.htm

Josh Boak. Trump pushes ahead on false claims Reflecting Pool was vandalized. Associated Press for *PBS NewsHour*, 6 August 2026. https://www.pbs.org/newshour/nation/trump-pushes-ahead-on-false-claims-reflecting-pool-was-vandalized

Mark F. Cancian and Chris H. Park. War costs make up a third of the $87.6 billion supplemental request. Center for Strategic and International Studies, 13 July 2026. https://www.csis.org/analysis/war-costs-make-third-876-billion-supplemental-request

Congressional Budget Office. *The Budget and Economic Outlook: 2026 to 2036.* Washington, DC, February 2026.

Congressional Budget Office. *Reconciliation Legislation of the Senate Committee on Homeland Security and Governmental Affairs.* Washington, DC, May 2026. https://www.cbo.gov/system/files/2026-05/HSGAC-and-Judiciary-Reconciliation.pdf

Riley Ceder. USS Gerald R. Ford breaks record for longest post-Vietnam deployment. *Navy Times*, 15 April 2026. Records the laundry fire of 12 March, which damaged a hundred berths, and the vacuum collection failures across some 650 heads. https://www.navytimes.com/news/your-navy/2026/04/15/uss-gerald-r-ford-breaks-record-for-longest-post-vietnam-deployment/

Nelson Cowan. The magical number 4 in short-term memory: A reconsideration of mental storage capacity. *Behavioral and Brain Sciences*, 24(1):87–114, 2001.

Committee for a Responsible Federal Budget. 90% of spending growth will come from health, retirement, veterans, and interest. Washington, DC, 12 February 2026. https://www.crfb.org/blogs/90-spending-growth-will-come-health-retirement-veterans-and-interest

Committee for a Responsible Federal Budget. Treasury auction yield hits highest in 25 years. Washington, DC, 14 August 2026. https://www.crfb.org/blogs/treasury-auction-yield-hits-highest-25-years

Clayton Thomas (coordinator), Christopher M. Blanchard, Jeremy M. Sharp, and Jim Zanotti. *U.S. Conflict with Iran.* CRS Report R48887, Congressional Research Service, Washington, DC, 26 March 2026. https://www.everycrsreport.com/reports/R48887.html

Congressional Research Service. *National Institutes of Health (NIH) Funding: FY1996–FY2026.* CRS Report R43341, Washington, DC, 2026.

Congressional Research Service. *U.S. Environmental Protection Agency FY2026 President's Budget Request.* CRS Report R48575, Washington, DC, 2026.

Seamus P. Daniels. Tracking FY 2027 defense appropriations, reconciliation, and the supplemental request. Center for Strategic and International Studies, 18 August 2026. https://www.csis.org/analysis/tracking-fy-2027-defense-appropriations-reconciliation-and-supplemental-request

Paul Dans and Steven Groves, editors. *Mandate for Leadership: The Conservative Promise.* The Heritage Foundation, Washington, DC, 2023.

Susan Davis and Lexie Schapitl. Sen. Mitch McConnell says he will not seek reelection in 2026. *NPR*, 20 February 2025. https://www.npr.org/2025/02/20/nx-s1-5304207/mitch-mcconnell-retirement-senate-kentucky-2026

European Central Bank. Global trade invoicing patterns: new insights. *The International Role of the Euro*, June 2026.

European Central Bank. *The International Role of the Euro*, June 2026. Frankfurt am Main, 2 June 2026. Gold 27 per cent of total official foreign reserves at end-2025 market prices against 22 per cent for United States Treasuries; 16 and 26 per cent respectively when valued at the end-2023 gold price. https://www.ecb.europa.eu/press/other-publications/ire/html/ecb.ire202606.en.html

Barry Eichengreen. *Exorbitant Privilege: The Rise and Fall of the Dollar and the Future of the International Monetary System.* Oxford University Press, 2011.

FXStreet Team. Breaking: Iran won't negotiate with the US until Trump's term is done. *FXStreet*, 10 August 2026. Reports the remark of an adviser to the parliamentary speaker, "we will accompany him until his term ends", dated to 20 January 2029. https://www.fxstreet.com/news/breaking-iran-wont-negotiate-with-the-us-until-trumps-term-is-done-202608101811

Chris Geidner. Jeanine Pirro's motion to dismiss the Reflecting Pool case is going to have ripple effects. *Law Dork*, 2 August 2026. Quotes the government's motion to dismiss filed in the Superior Court for the District of Columbia, 31 July 2026. https://www.lawdork.com/p/jeanine-pirro-motion-to-dismiss-reflecting-pool

Friedrich Goltz. *Beiträge zur Lehre von den Functionen der Nervencentren des Frosches*. August Hirschwald, Berlin, 1869. The heated-water experiments are at pp. 127–130. https://archive.org/details/beitrgezurlehre01goltgoog

Sam Gringlas. For Mitch McConnell and Congress, health transparency is a choice, not a requirement. *NPR*, 11 July 2026. https://www.npr.org/2026/07/10/nx-s1-5885905/as-sen-mitch-mcconnell-remains-hospitalized-calls-for-transparency-grow

Derek Halpenny. FX focus: FX reserve managers move into USD in Q1 but will this be sustained? MUFG Research, 14 July 2026. https://www.mufgresearch.com/fx/fx-focus-fx-reserve-managers-move-into-usd-in-q1-but-will-this-be-sustained-14th-july-2026/

Friedrich A. Hayek. The use of knowledge in society. *American Economic Review*, 35(4):519–530, 1945.

Institute of International Finance. *Global Debt Monitor*. Washington, DC, February 2026.

International Monetary Fund. *Currency Composition of Official Foreign Exchange Reserves (COFER)*, 2026Q1 release. Washington, DC, 2026. Dollar share 57.13 per cent of $13.10 trillion in total reserves. https://data.imf.org/en/datasets/IMF.STA:COFER

Timothy Inklebarger. Trump purge at FDA and USDA triggers food safety 'brain drain'. *FoodNavigator*, 16 February 2026. Figures from the Office of Personnel Management's federal workforce data: 3,859 Food and Drug Administration staff lost in 2025 and 473 in 2026 to date, leaving 16,602. https://www.foodnavigator.com/Article/2026/02/16/fda-and-usda-staff-cuts-under-trump-raise-food-safety-risks/

Iran International. Araghchi says no decision yet to resume talks with US. 14 August 2026. Messages passed through Qatar and Pakistan are stated not to be negotiations. https://www.iranintl.com/en/202608145480

International Union of Geological Sciences and International Commission on Stratigraphy. The Anthropocene: IUGS–ICS statement. 20 March 2024. Subcommission on Quaternary Stratigraphy vote: 4 for, 12 against, 3 abstentions. https://www.iugs.org/the-anthropocene-iugs-ics-statement/

Lara Korte. Iran war has consumed nearly 67% of US Patriot interceptor stockpile, report says. *Stars and Stripes*, 30 July 2026, reporting analysis by the Center for Strategic and International Studies. https://www.stripes.com/theaters/middle_east/2026-07-30/iran-war-missile-stockpiles-patriots-22406307.html

Anandi Mani, Sendhil Mullainathan, Eldar Shafir, and Jiaying Zhao. Poverty impedes cognitive function. *Science*, 341(6149):976–980, 2013.

Caleb McCullough. Social media users spread conspiracy theories about Mitch McConnell after photo release. *PolitiFact*, 13 July 2026. https://politifact.com/factchecks/2026/jul/13/tweets/mitch-mcconnell-rehabilitation-photo-ai-reused/

Caleb McCullough. The internet says Mitch McConnell's hospital photo is fake. The evidence says otherwise. *Poynter*, 14 July 2026. https://www.poynter.org/fact-checking/2026/is-mitch-mcconnell-photo-real-fake-newspaper/

Mary Ellen McIntire and Niels Lesniewski. McConnell to step down from Senate leadership. *Roll Call*, 28 February 2024. Announced 28 February 2024; the successor took the post on 3 January 2025. https://rollcall.com/2024/02/28/mcconnell-to-step-down-from-senate-leadership/

George A. Miller. The magical number seven, plus or minus two: Some limits on our capacity for processing information. *Psychological Review*, 63(2):81–97, 1956.

Associated Press for *NPR*. McConnell says a fall led to his hospitalization, breaking weeks of silence. 13 July 2026. https://www.npr.org/2026/07/13/nx-s1-5891492/mcconnell-fall-hospitalization

OECD. Affordable Housing Database, indicator HM1.2: Housing prices. OECD, Paris, 2025.

Natalie Oliverio. Senator presses Navy for answers on USS *Abraham Lincoln* deployment. *Military Times*, 13 August 2026. https://www.militarytimes.com/news/your-military/2026/08/13/senator-presses-navy-for-answers-on-uss-abraham-lincoln-deployment/

Partnership for Public Service. *The Federal Workforce One Year into the Trump Administration.* Washington, DC, 16 April 2026.

Paul Preston. *Franco: A Biography.* HarperCollins, London, 1993.

Radio Free Europe/Radio Liberty. Dead silence: delays, deception mark the deaths of dictators. 2 September 2016. https://www.rferl.org/a/karimov-death-other-dictators-mystery-deception-/27964175.html

Greg Ritchie. U.S. set to pay most for 30-year debt in quarter of a century. *Fortune* (Bloomberg), 13 August 2026. https://fortune.com/2026/08/13/us-debt-treasury-30-year-highest-25-years/

Rebecca Schneid. Iran's president signals willingness to resume peace talks. What stands in the way? *TIME*, 1 September 2026. https://time.com/article/2026/09/01/irans-president-willing-to-resume-peace-talks-what-stands-in-the-way-/

W. T. Sedgwick. On variations of reflex-excitability in the frog, induced by changes of temperature. *Studies from the Biological Laboratory of the Johns Hopkins University*, II:385–410. N. Murray, Baltimore, July 1888. https://archive.org/details/studiesfrombiol00martgoog

Robert Service. *Stalin: A Biography.* Harvard University Press, Cambridge, MA, 2004.

Securities Industry and Financial Markets Association. *SIFMA Capital Markets Fact Book, 2026 Edition.* New York, 13 August 2026. United States equities 43.7 per cent of $157.8 trillion in global equity market capitalisation at end-2025, 4.4 times the next largest market. https://www.sifma.org/wp-content/uploads/2025/07/SIFMA-Capital-Markets-Fact-Book-2026-Edition.pdf

Nicholas Slayton. Sailors on the USS *Abraham Lincoln* reportedly dealing with poor conditions, weak morale, families say. *Task & Purpose*, 8 August 2026. https://taskandpurpose.com/news/uss-abraham-lincoln-conditions-sailors/

Randy J. Stine. FCC 2026 budget proposal totals $416.1 million. *Radio World*, 3 June 2025. Quotes the commission's budget submission on continued workforce reduction through retirements, the deferred-resignation route and other attrition. https://www.radioworld.com/news-and-business/headlines/fcc-2026-budget-proposal-totals-416-1-million

SWIFT. *RMB Tracker and Global Currency Tracker*, July 2026. La Hulpe, 2026.

Rachel Treisman. Ripple effects: three people have Reflecting Pool charges dropped after Hearn request. *NPR*, 3 August 2026. https://www.npr.org/2026/08/03/nx-s1-5918001/reflecting-pool-vandalism-charges-dropped

V-Dem Institute. *Democracy Report 2026: Unraveling the Democratic Era?* Tenth edition. University of Gothenburg, 2026.

Philip Wang. Lincoln Memorial Reflecting Pool repairs awarded to same contractor. *TIME*, 6 July 2026. https://time.com/article/2026/07/06/lincoln-memorial-reflecting-pool-repairs-Doug-Burgum-Algae-Gash/

Howard Wilkinson. Analysis: McConnell voted against 3 Trump cabinet picks. Does it matter? *WVXU*, 14 February 2025. https://www.wvxu.org/politics/2025-02-14/analysis-mitch-mcconnell-trump-confirmations

Chenyu Zhou, Huacan Chai, Wenteng Chen, Zihan Guo, Rong Shan, Yuanyi Song, Tianyi Xu, Yingxuan Yang, Aofan Yu, Weiming Zhang, Congming Zheng, Jiachen Zhu, Zeyu Zheng, Zhuosheng Zhang, Xingyu Lou, Changwang Zhang, Zhihui Fu, Jun Wang, Weiwen Liu, Jianghao Lin, and Weinan Zhang. Externalization in LLM agents: A unified review of memory, skills, protocols and harness engineering. arXiv:2604.08224, 2026.